



UNIT 7 & 8: Part 2

Supply and Demand and Competitive Equilibrium

Introduction

Pineapple prices explode as South Africans rush to brew their own beer

Business Insider SA
May 11, 2020, 01:18 PM



Pineapple prices have skyrocketed as South Africans – desperate for booze during lockdown – turn their hand to amateur brewing.

According to numbers from the Johannesburg market, pineapple prices have more than tripled between April 2019, when it traded at an average price of around R4.96/kg, and April 2020, when it hit R15.72/kg.

In April, pineapples worth more than R14 million were sold at the Johannesburg market - double the amount sold in March.

Joe Mazibuko, the acting CEO of the Joburg market, notes that pineapple prices since the start of the year are 82% higher than in the same period last year, due in part to strong demand from retailers.

“It is suspected that most pineapples went towards the brewing of homemade beer in the face of the closure of bottle stores.”



8.1

Mark Harris of Langholm Farms in Bathurst, the largest pineapple producer in South Africa, says demand from Western Europe for South African pineapples has been good in recent weeks. Anecdotal reports claim that the coronavirus pandemic has bolstered sales of fruit in parts of Europe as consumers go on a health kick amid the pandemic.

UNIT 7 & 8: Part 2 Overview

- Demand and Supply
- Competitive market equilibrium
- Price taking firms
- Changes in supply and demand
- Competitive market efficiency

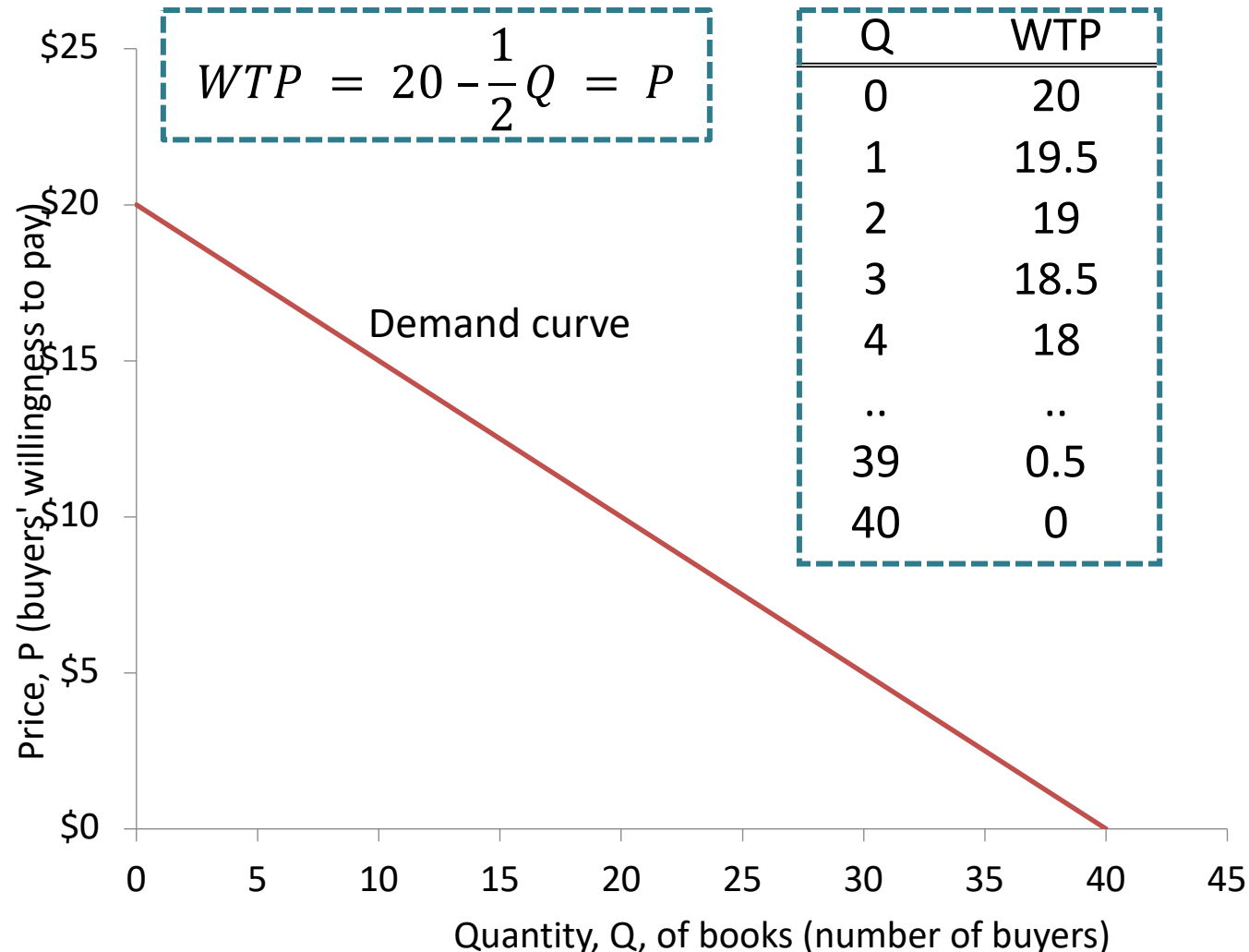
8.1 Demand and Supply

Consider a market for a second-hand economics textbooks

- Demand comes from students who are about to take the course and they will differ in their **willingness to pay (WTP)**
 - No one would be willing to pay more than the price of a new copy for a secondhand textbook
- Supply comes from students who have previously completed the course, who will differ in their **willingness to accept (WTA)**
 - That is their reservation price – lowest price at which someone is willing to sell a good



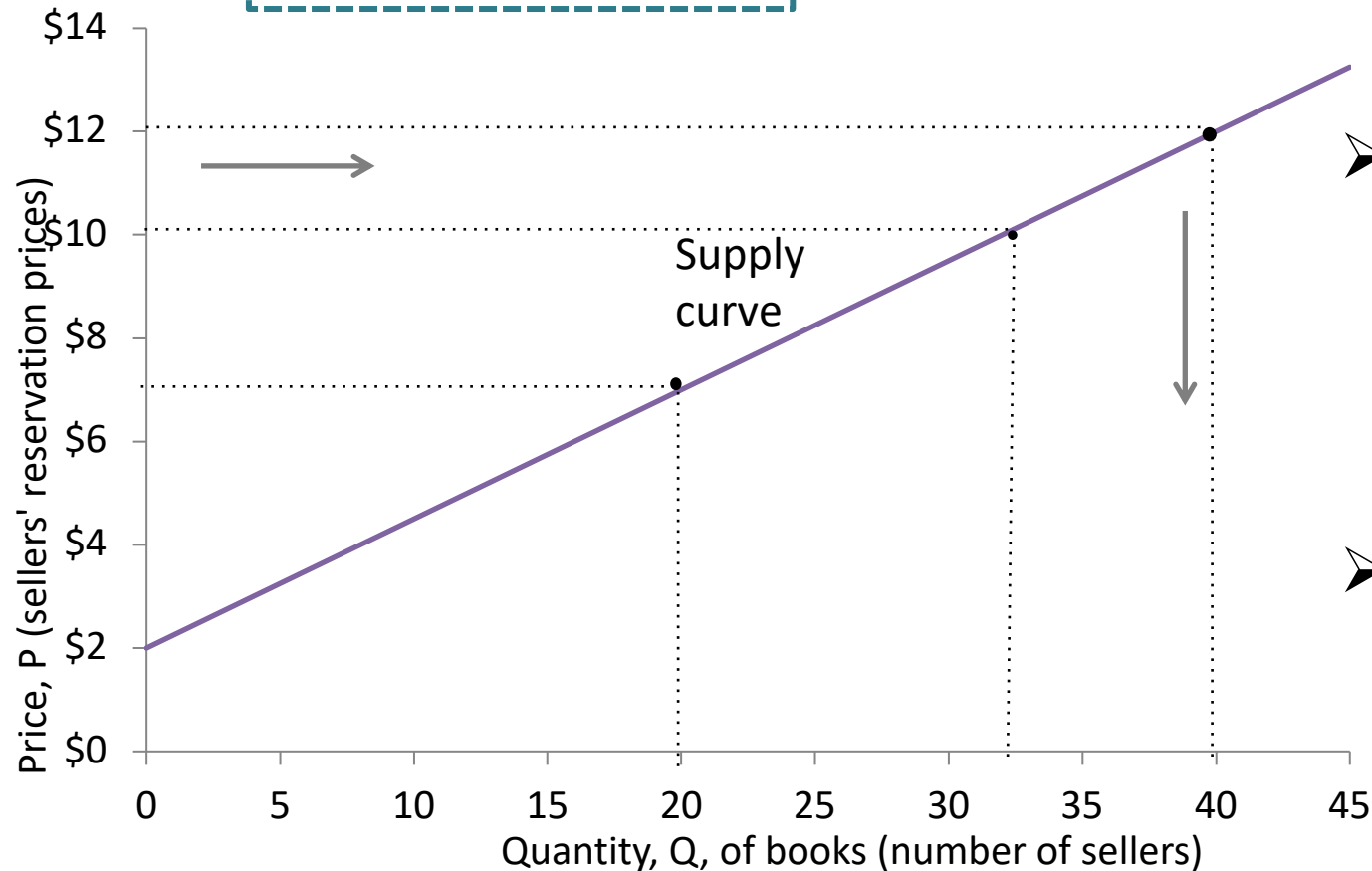
Demand for used textbooks



- Demand curve represents the WTP of buyers
 - i.e. number of students willing to buy at a given price
- Students' **willingness to pay** depends on:
 - how interested they are in the subject
 - how useful they think the textbook will be
 - how much money they (or their parents) have
- Demand can also be represented as
 - a table
 - an equation

Supply of used textbooks

$$WTA = 2 + \frac{1}{4}Q = P$$



➤ Supply curve is drawn by lining sellers from those with the lowest *reservation price*

➤ WTA

- First seller has reservation price (WTA) of \$2
- 20th seller is WTA \$7
- 32nd seller is WTA \$10
- 40th seller is WTA \$12
- This produces the supply curve

➤ Supply can also be represented as

- an equation
- a table

8.2 The market and the equilibrium price

Markets (and their institutions) bring many buyers and sellers together:



Grand Bazaar, Istanbul



Chicago Mercantile Exchange

LOOK INSIDE! **Principles of Economics, 7th Edition (Hardco**
by N. Gregory Mankiw

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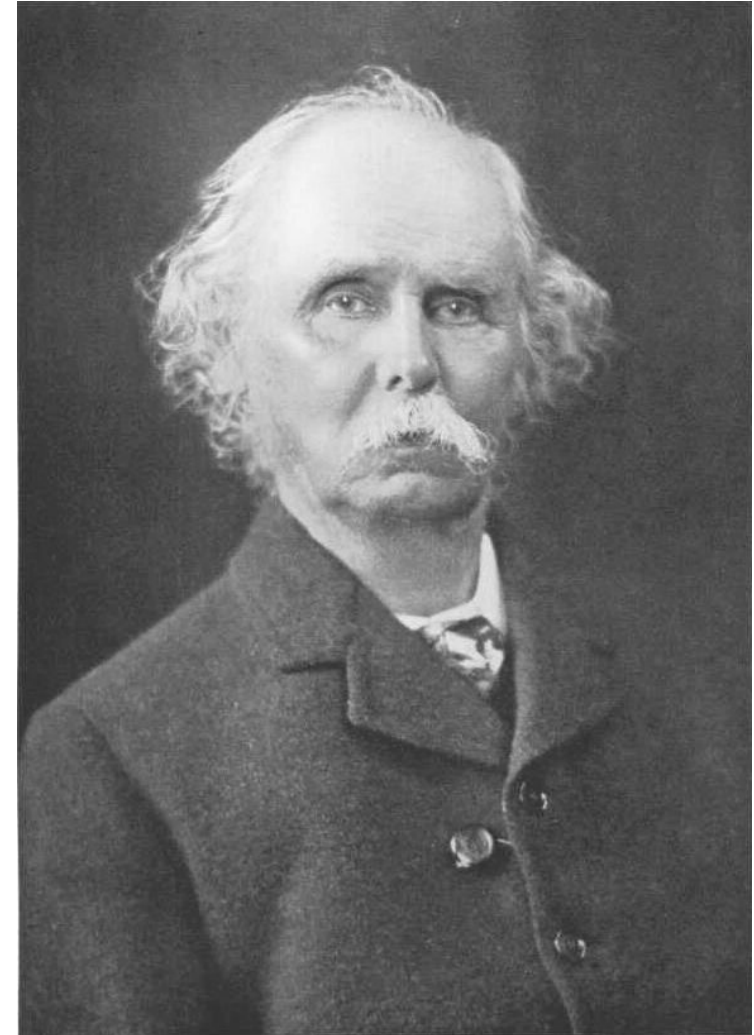
Condition
☐ New
☐ Rental
☒ Used
☐ Like New
☒ Very Good
☐ Good
☐ Acceptable

Price + Shipping	Condition (Learn more)
\$54.65 ✓prime & FREE Shipping Details	Used - Very Good YES! BOOK READY TO SHIP WITH FREE TRACKING NUMBER. NO HI-LIGHTING. ✓ Read more
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Amazon.com

Alfred Marshall (1842-1924)

- Marshall introduced his model of supply and demand at the end of the 19th century using an example similar to our market for secondhand textbooks
- Grain exchange
 - Marshall describes how the supply curve would be determined by prices farmers would be **willing to accept**, and demand curve by **willingness to pay** by merchants
 - Marshall called the price that would equate quantity supplied and quantity demanded the ***equilibrium price***
 - If a price was above equilibrium price then farmers would want to sell more BUT fewer merchants would want to buy at the higher price → resulting in an ***excess supply***. Eventually the price will adjust downwards so a new equilibrium is reached.



Demand and quantity demanded

- Marshall described how the demand curve would be determined by prices buyers would be **willing to pay**
- That is, quantity demanded is a function of price

$$Q_d = f(P), \text{ e.g:}$$

$$Q_d = 40 - 2P$$

- Rearranging this function so that $P = f(Q_d)$ gives us the **inverse demand function**, which when plotted gives us the demand curve

$$Q_d = 40 - 2P$$

$$2P = 40 - Q_d$$

$$P = 20 - \frac{1}{2}Q_d$$

Supply and quantity supplied

- Similarly, Marshall described how the supply curve would be determined by prices sellers would be **willing to accept**
- That is, quantity supplied is a function of price

$Q_s = f(P)$, e.g:

$$Q_s = 4P - 8$$

- Rearranging this function so that $P = f(Q_s)$ gives us the **inverse supply function**, which when plotted gives us the supply curve

$$Q_s = 4P - 8$$

$$4P = 8 + Q_s$$

$$P = 2 + \frac{1}{4}Q_s$$

Competitive equilibrium

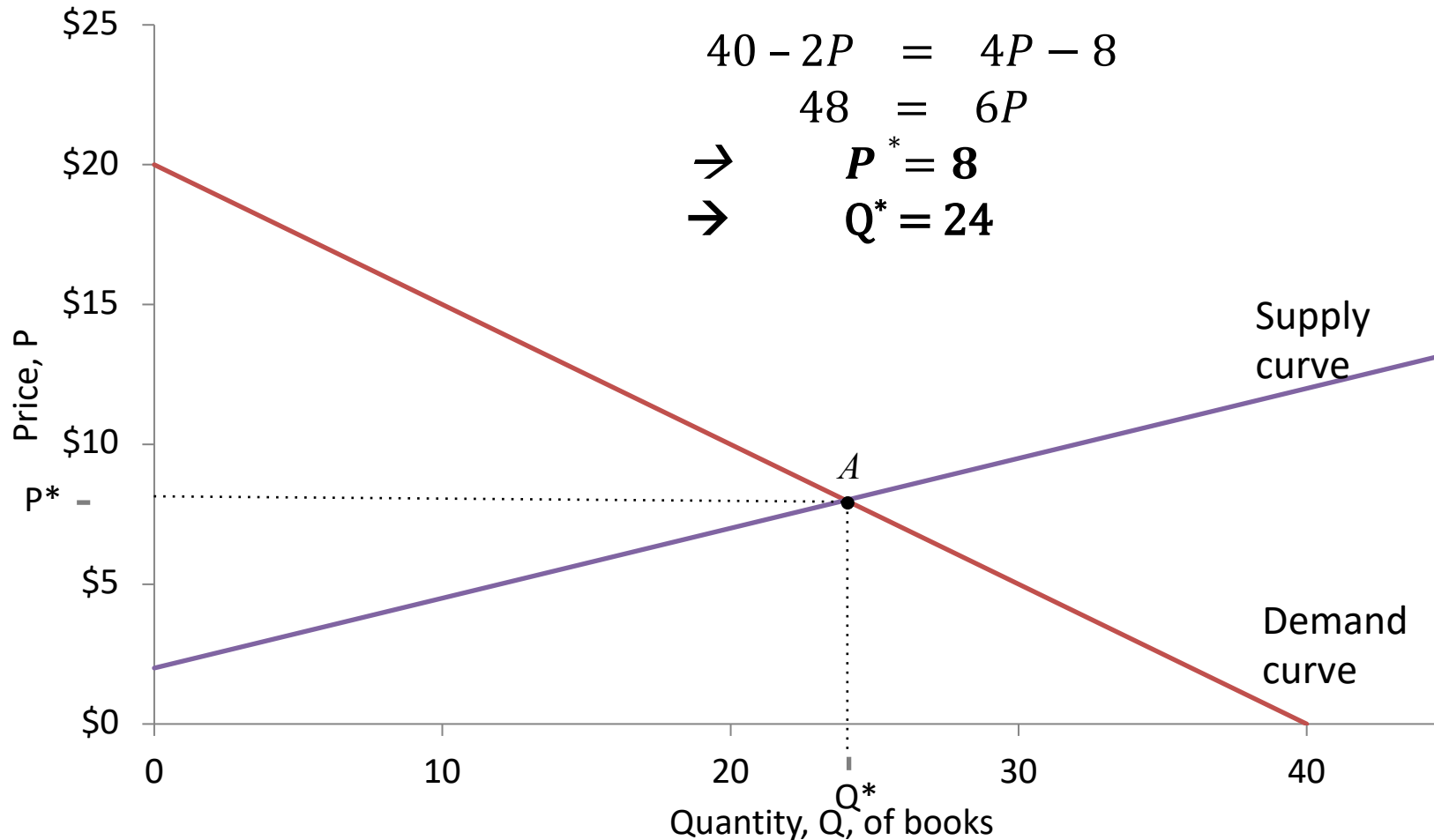
$$Q_d = Q_s$$

$$40 - 2P = 4P - 8$$

$$48 = 6P$$

$$\rightarrow P^* = 8$$

$$\rightarrow Q^* = 24$$



- The price (P^*) that equates quantity supplied and quantity demanded is the *equilibrium (market clearing) price*
- All buyers and sellers can execute their planned sales / purchases

Price taking

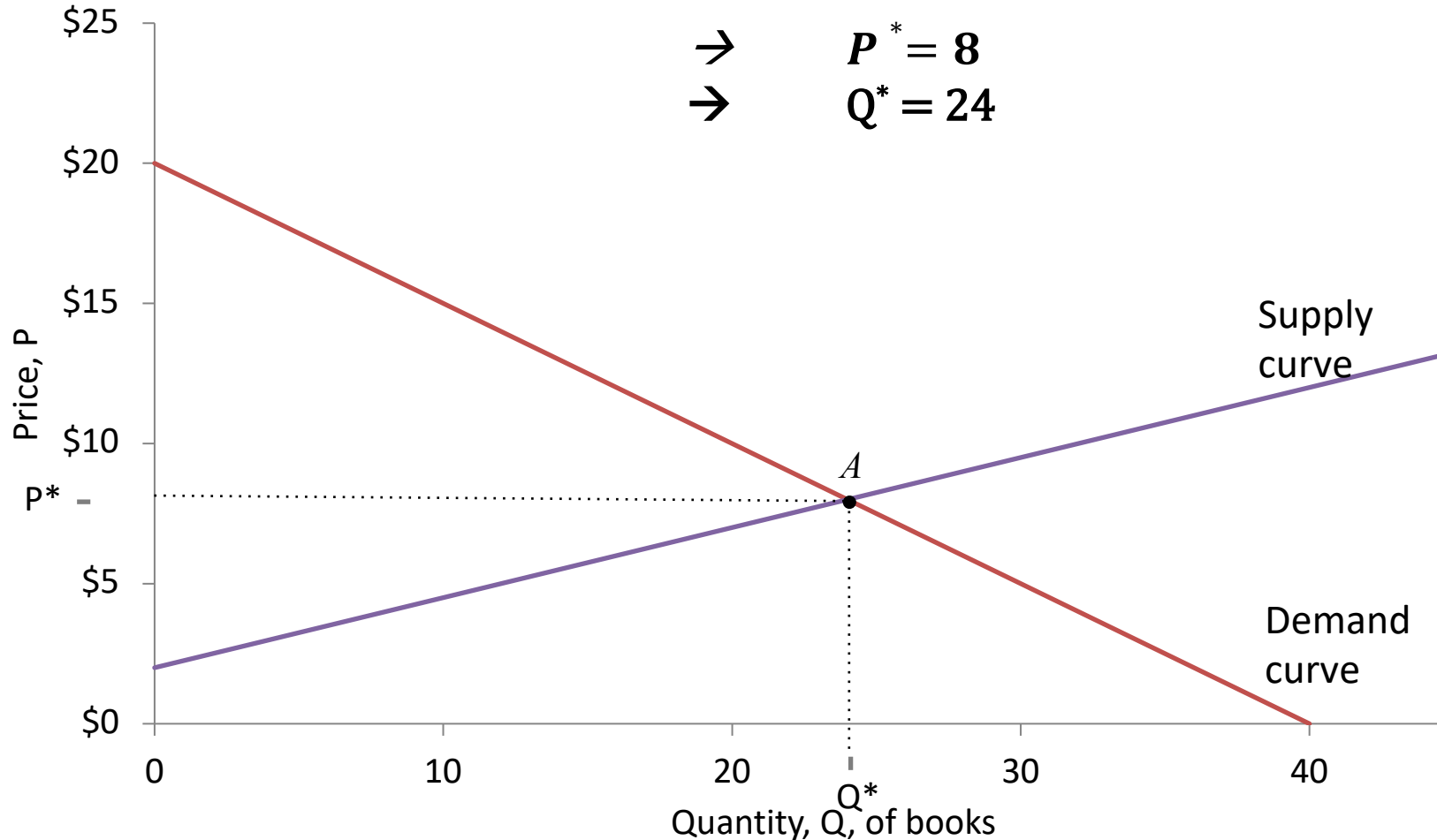
- Why would students in a secondhand market for a first-year economics textbook have to accept the prevailing market price?
- Let's consider some important characteristics of this market:
 - The product is identical (the same textbook)
 - There are many buyers and many sellers (each year, there are at least 1700 first-year students)
 - Each seller has only a small quantity of books to sell (in this case, one), and each buyer is only looking to buy a small quantity of books (in this case, one)
 - There is (almost) perfect information (sellers and buyers are well informed on the prices that students are WTA and WTP)
- If a seller asked a higher price, the buyer would look for someone with a lower WTA
 - Why would a seller not ask a lower price than the prevailing market price?
- If a buyer offered a lower price, the seller would look for someone with a higher WTP

Competitive equilibrium

$$Q_d = Q_s$$

$$\rightarrow P^* = 8$$

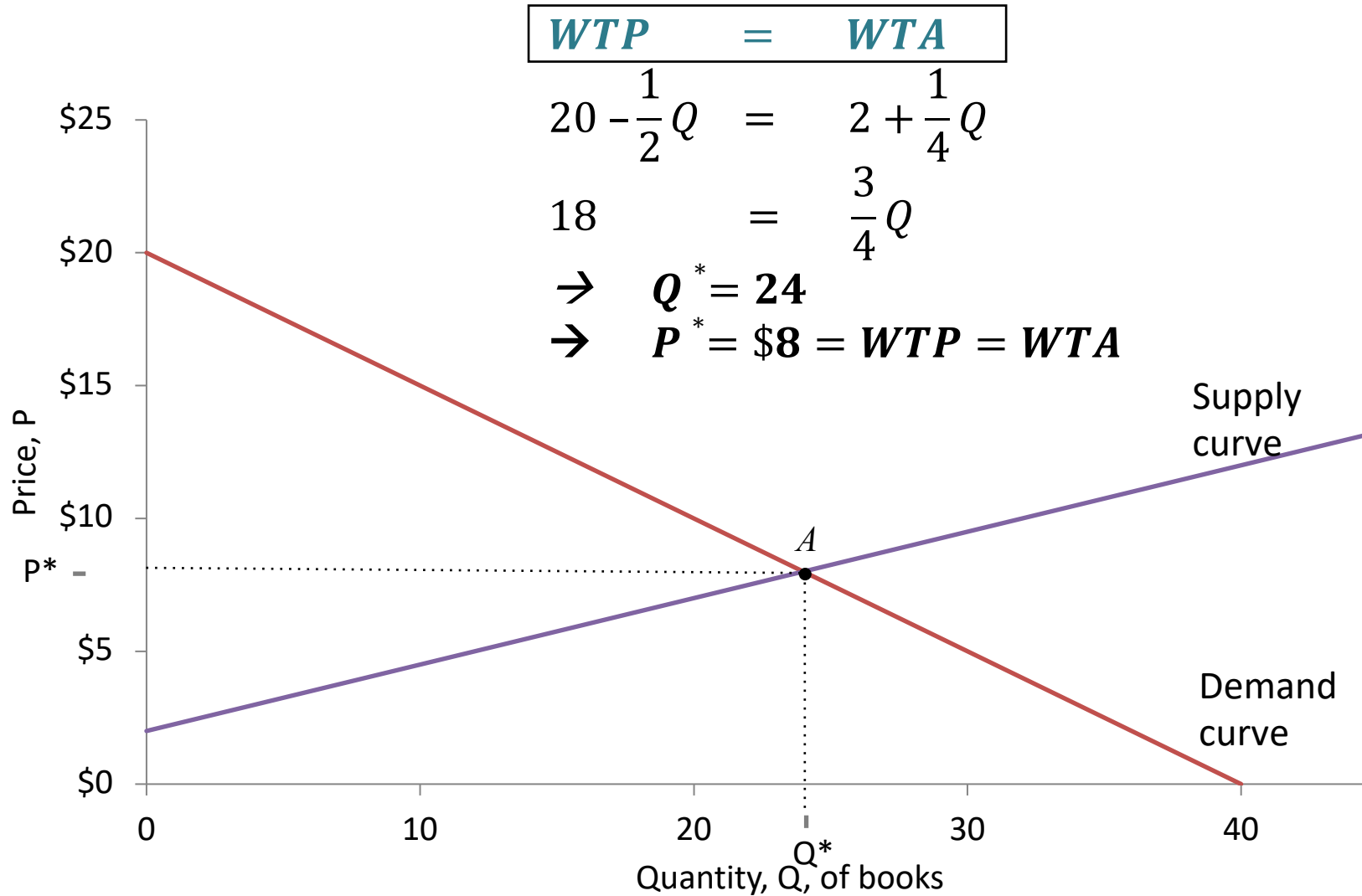
$$\rightarrow Q^* = 24$$



- The market equilibrium is a *Nash equilibrium*:
 - Selling at a price higher than $P^* \rightarrow$ zero sales
 - Selling at a price lower than $P^* \rightarrow$ no benefit
 - Buyer offering to pay less than $P^* \rightarrow$ no seller will agree to sell
 - Buyer offering to pay more than $P^* \rightarrow$ no benefit

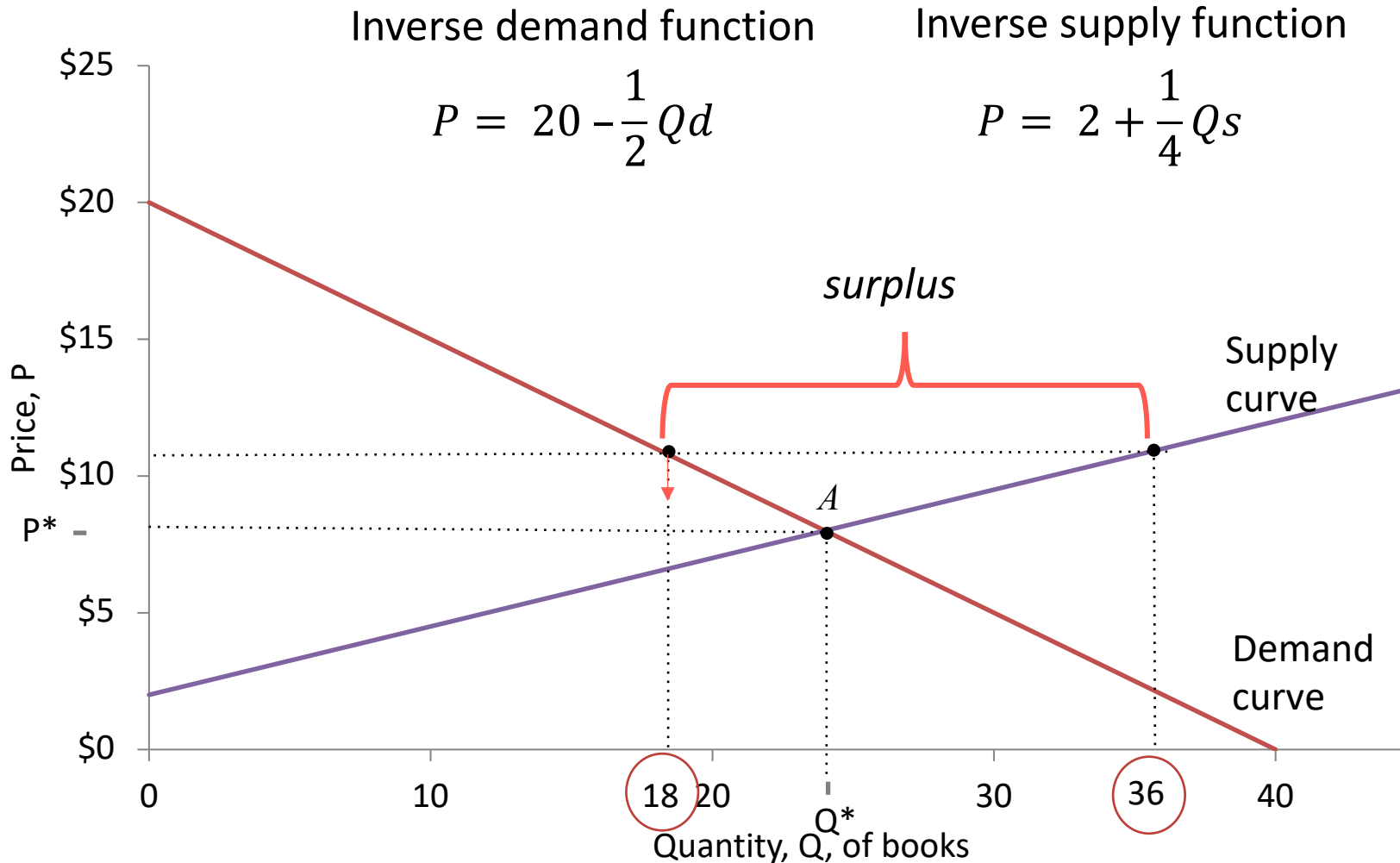
No one has an incentive to change behaviour
- All buyers and sellers are price takers
- P^* is the market clearing price.

Competitive equilibrium



- Alternative way to finding equilibrium: equating inverse demand and supply functions.
- Clearly this will give the same P^* and Q^* as obtained from setting $Q_d = Q_s$.
- At the margin, the buyer and the seller of the last unit traded have equal WTP and WTA.
- How does the market price adjust to P^* ?

Competitive equilibrium



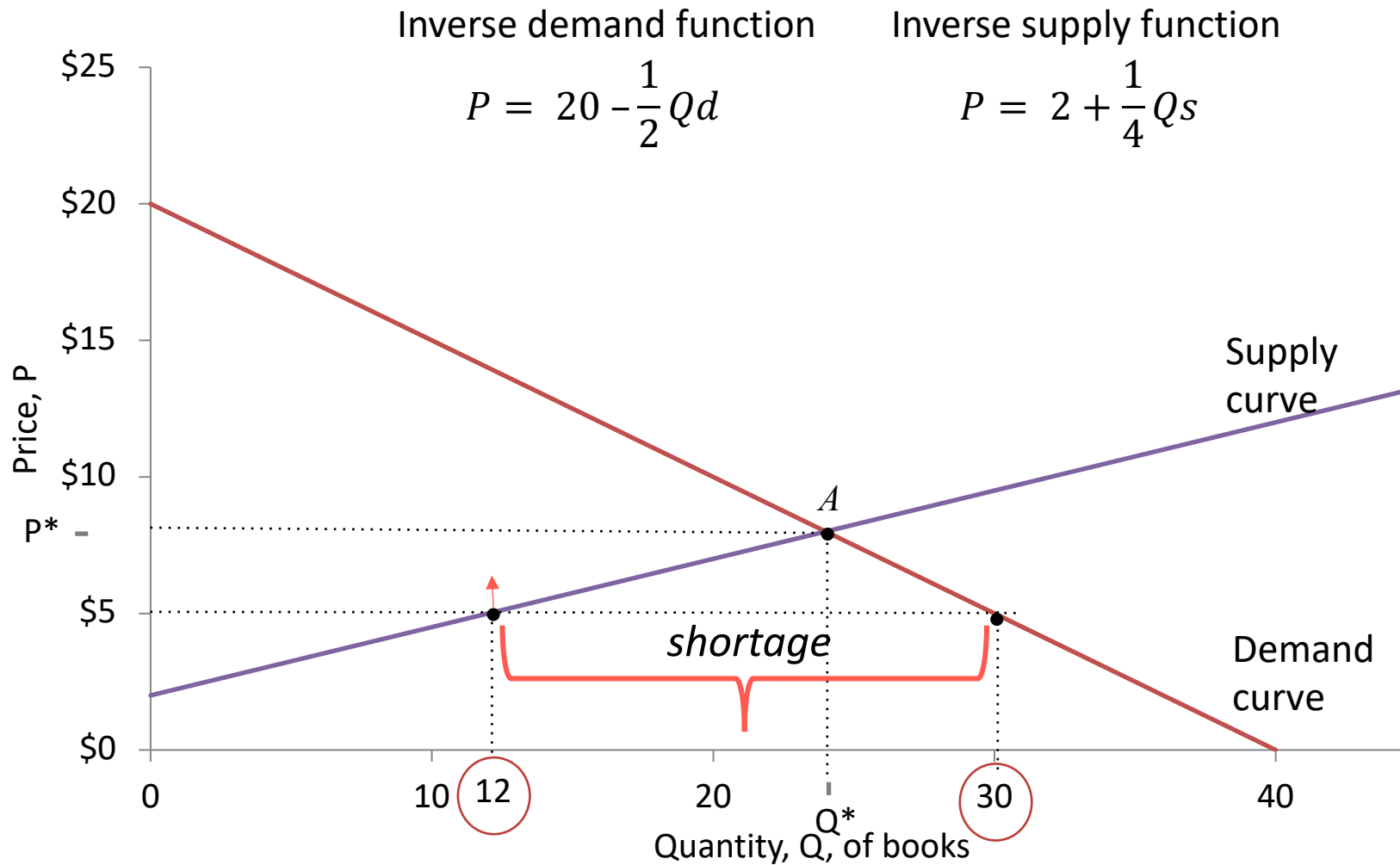
If the price is too high, there is *excess supply (surplus)*

e.g. if $P = \$11$, $Q_d = 18$

$Q_s = 36$

- There is a surplus of $36 - 18 = 18$ units: some sellers have unsold stock
- Only 18 units are bought and sold
- $P = WTP > WTA$ (at $Q=18$)
- Some sellers and buyers would both agree to price discount
- This pushes price down

Competitive equilibrium



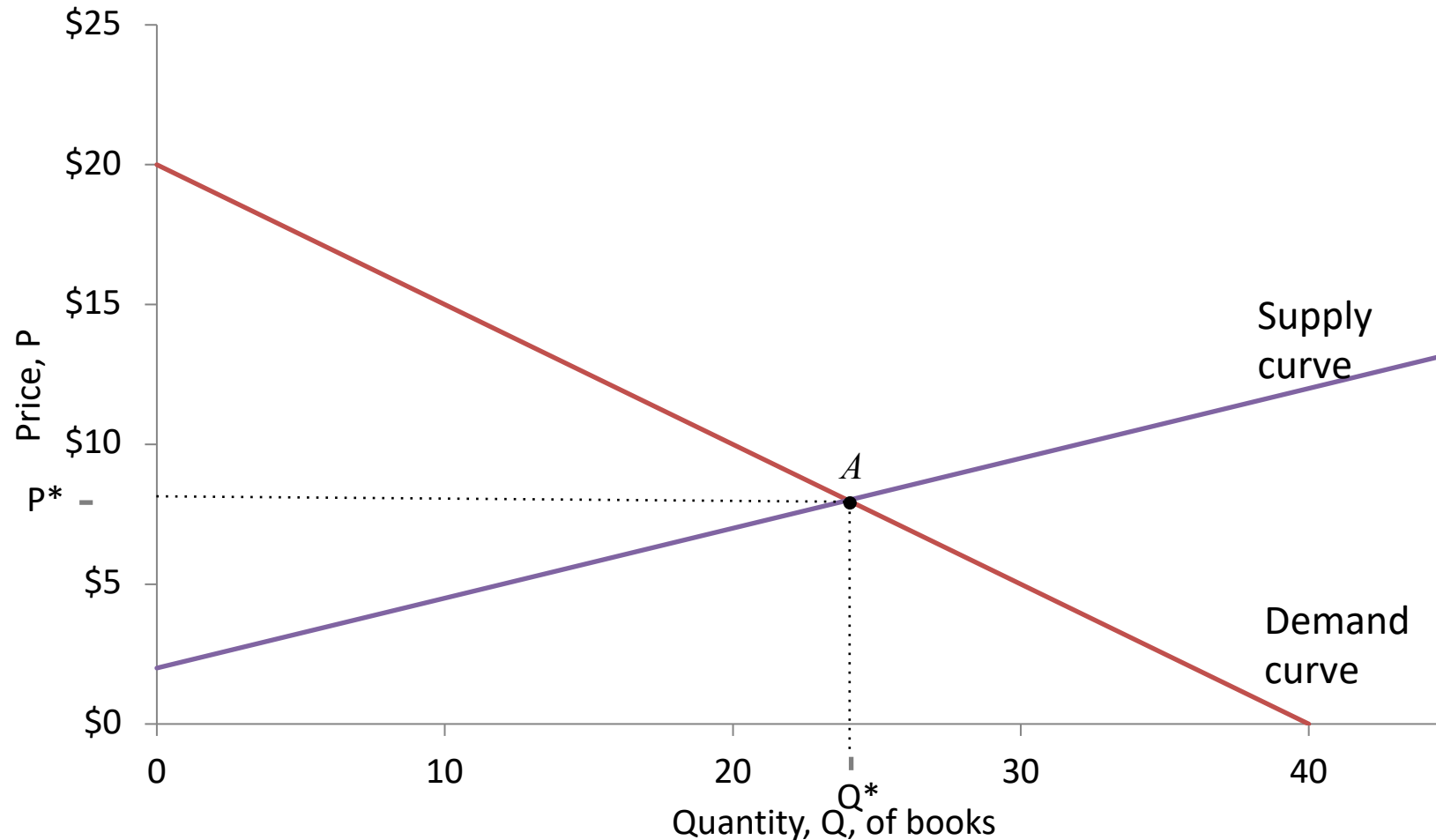
If the price is too low there is *excess demand (shortage)*

e.g. if $P = \$5$, $Q_d = 30$

$Q_s = 12$

- There is a shortage of $30 - 12 = 18$ units: some buyers cannot find stock
- Only 12 units are bought and sold
- $WTP > WTA = P$ (at $Q = 12$)
- Some sellers and buyers would both agree to price increase
- This pushes price up

Competitive equilibrium



Competition (many buyers and sellers selling identical products) drives the price towards this *competitive equilibrium*:

The *market clears* at

$$P^* = \$8$$

$$Q^* = 24 \text{ (where } Q_d = Q_s \text{)}$$

All sellers can easily sell the number of units they want to → no reason to lower price

All buyers can buy the number of units they want to → no reason to offer a higher price

Exercise

Inverse demand is $P = 600 - \frac{1}{3}Q_d$ and inverse supply is $P = 100 + Q_s$.

1. What are the demand and supply functions? ($Q_d = \dots$, $Q_s = \dots$)
2. What are the equilibrium price P^* and quantity Q^* ?
3. Describe the situation in this market if the price is 300.
4. Draw a graph to illustrate the situation.

8.8 The model of Perfect competition

- In perfect competition both buyers and sellers are **price takers**.
- For a market to be perfectly competitive:
 - there must be many buyers
 - there must be many sellers
 - the product must be homogeneous (or undifferentiated)
 - buyers and sellers must know the market price and all other relevant information
- In the long run there are free entry and exit



Is the pizza market perfectly competitive?

Market for soybeans



The market for soybeans is perfectly competitive.



If you want to buy or sell soybeans, the competitive equilibrium price is \$8.68 a bushel.

Commodity markets



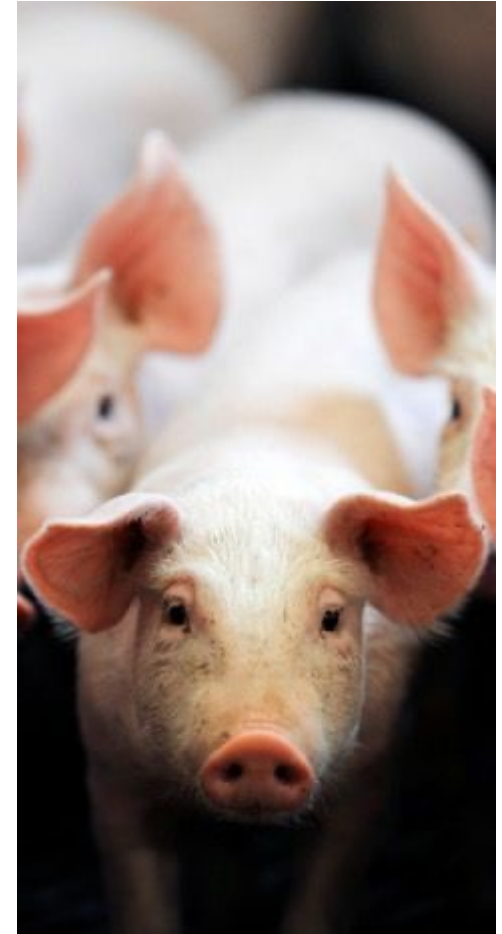
Light crude oil is selling for \$62/barrel.



Unleaded gasoline is selling for \$1.65/gallon.



Copper is selling for \$2.74/pound.



Lean hogs are selling for \$0.55/pound.

Price setters vs. price takers

- The model of perfect competition describes an idealized market structure.
- Even if some conditions for perfect competition are not met, there may be enough competition that we can assume firms are price takers.
- A simplified model can provide useful predictions when its assumptions are only approximately true.
- For example, the model of supply and demand can be useful for understanding markets of somewhat differentiated products.



Check out Hasan Minhaj's *Patriot Act* episode on Amazon.com on Netflix or [Youtube](#).

Is Amazon.com a price setter or price taker?

8.8 The model of Perfect competition

➤ In a perfectly competitive market:

- All transactions occur at a **single price**.
- At that price, the quantity supplied equals the quantity demanded and the market clears.
- There are **no shortages / surplus** in equilibrium.
- All buyers and sellers can execute their plans.
- There is no need to search around for better quality or prices.
- The result is Pareto efficient (see Part 3: Efficiency and fairness)

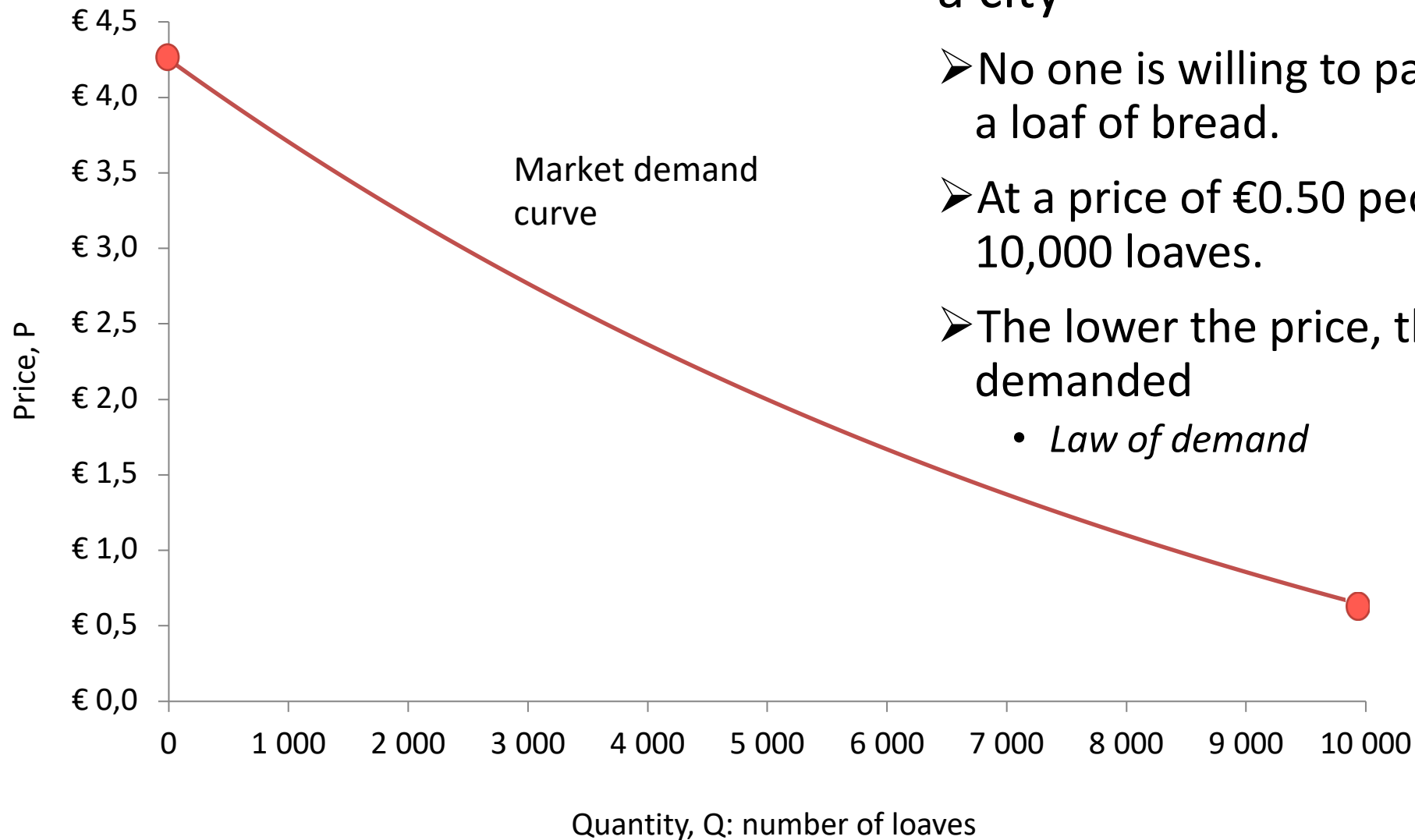
8.8 The model of Perfect competition

- Competition eliminates bargaining power:
 - There are many buyers and sellers.
 - Each buyer and seller is a small player in the market.
 - No buyer or seller has power to change the market price.
 - At the margin there is no rent to bargain over because $WTA = WTP$



8.3 Price taking firms

8.3 Price-taking firms



Consider the daily demand for bread in a city

- No one is willing to pay more than €4.25 for a loaf of bread.
- At a price of €0.50 people would demand 10,000 loaves.
- The lower the price, the greater the quantity demanded
 - *Law of demand*

Price-taking firms

- *Price-setting* firms choose the price-quantity combination that maximizes profits.
- *Price-taking* firms take the price as given and choose the quantity that maximizes profits.
- The competitive market model assumes that a firm in a competitive market
 - can sell any quantity it wants to at the market price
 - will not influence the market price

Price-taking firms

➤ Suppose you are the owner of a small bakery with:

- a *fixed cost* (e.g. rent of premises)
- rising *marginal costs* (MC)
- *Average costs* (AC) that fall and then rise (U-shaped)

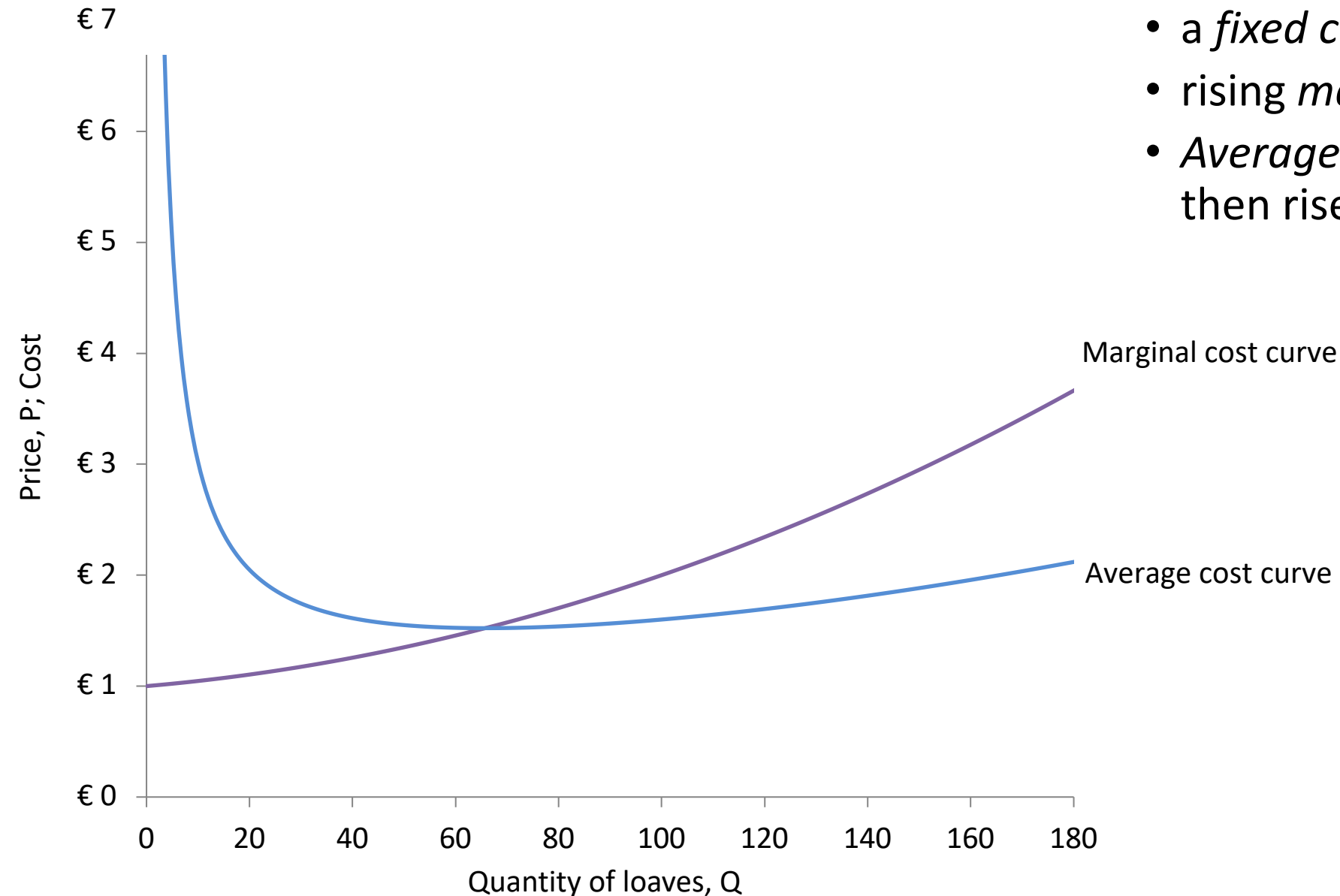
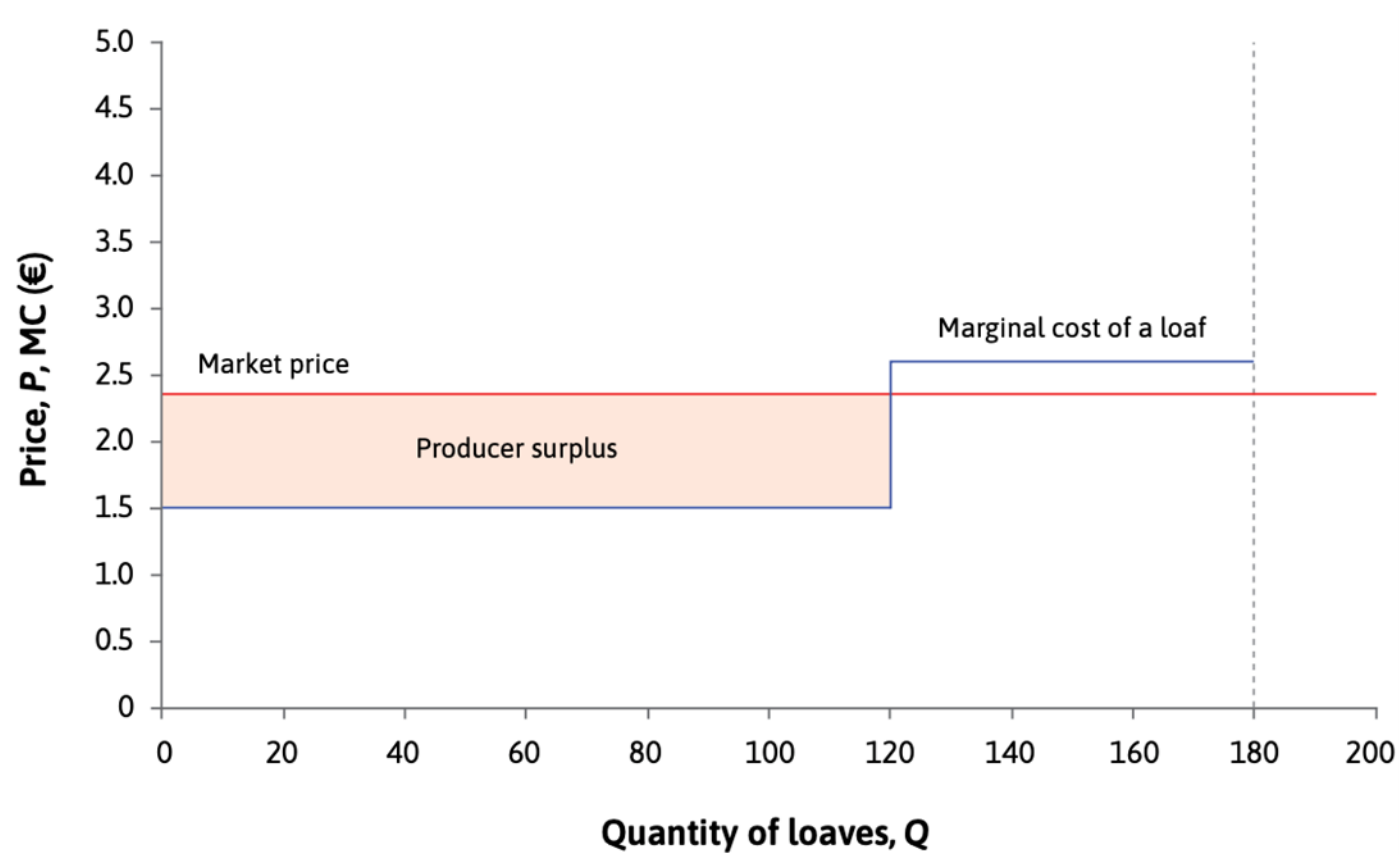


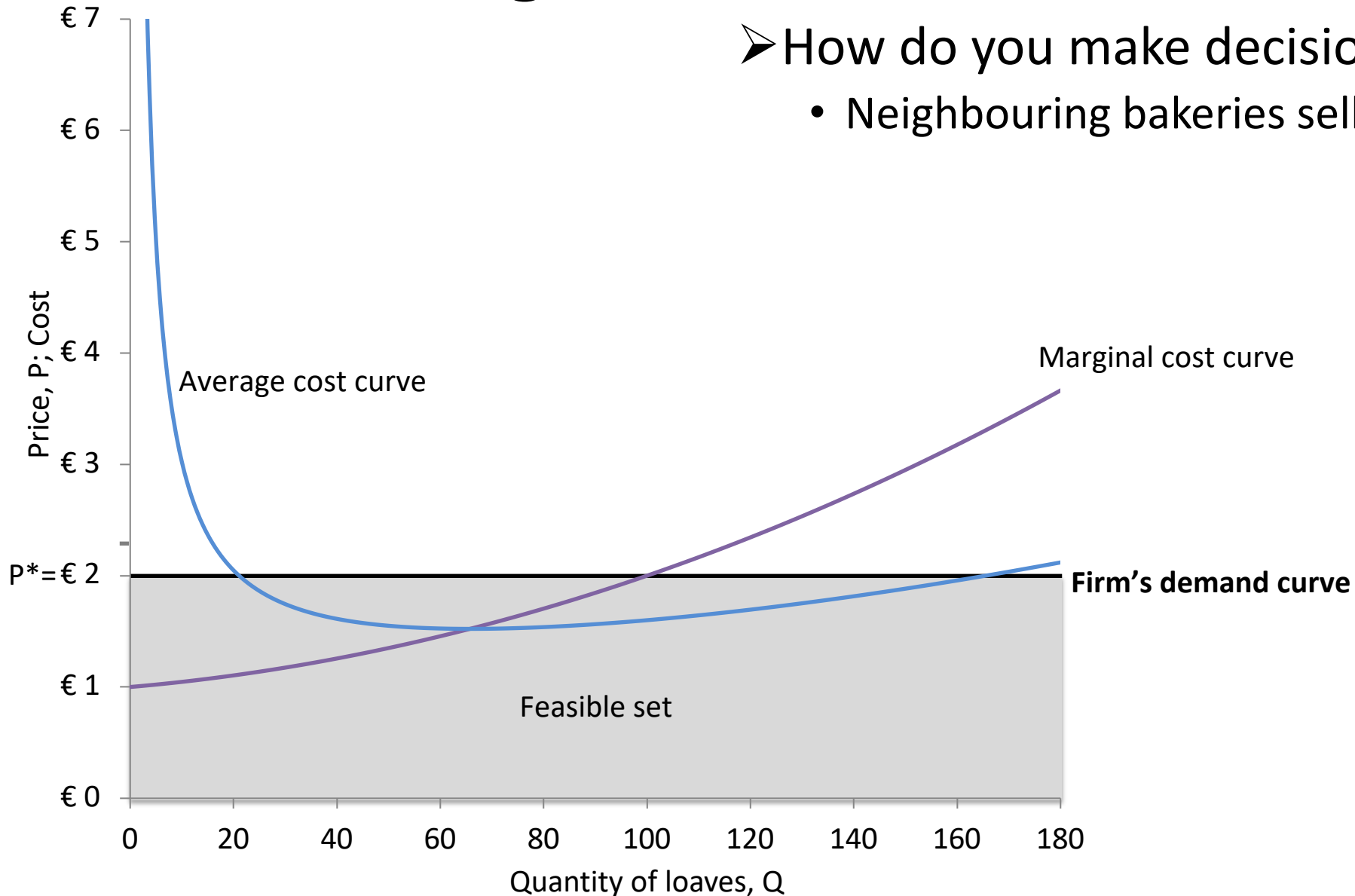
Fig 8.8



Price-taking firms

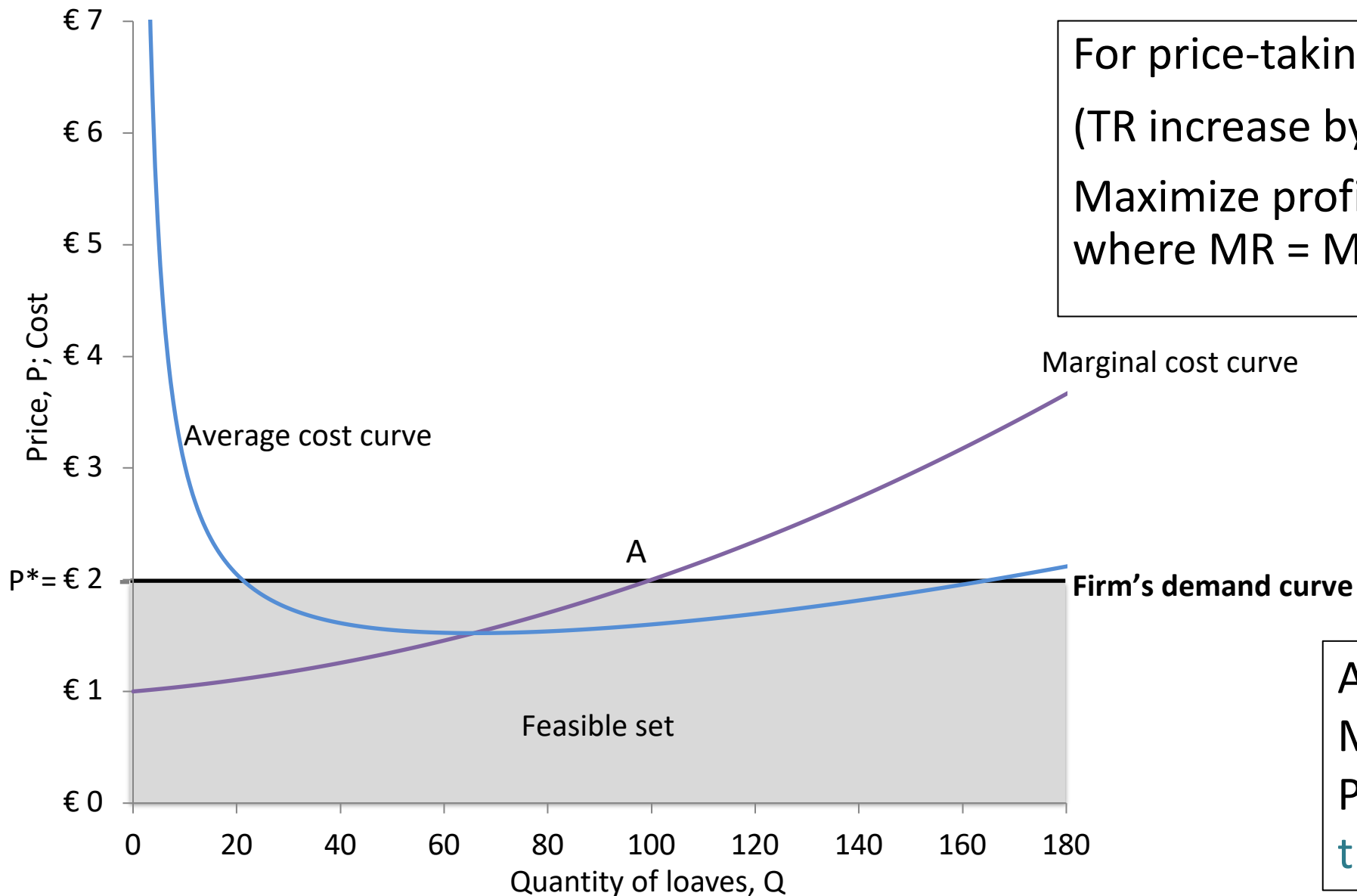
➤ How do you make decision on P and Q ?

- Neighbouring bakeries sell identical loaves for €2.00



- Your demand curve is a straight horizontal line at €2.00
- You cannot charge a higher price.
- You could charge a lower price (would you want to?)
- *price taker* → Feasible set is the shaded area

Price-taking firms: Marginal Revenue (MR)

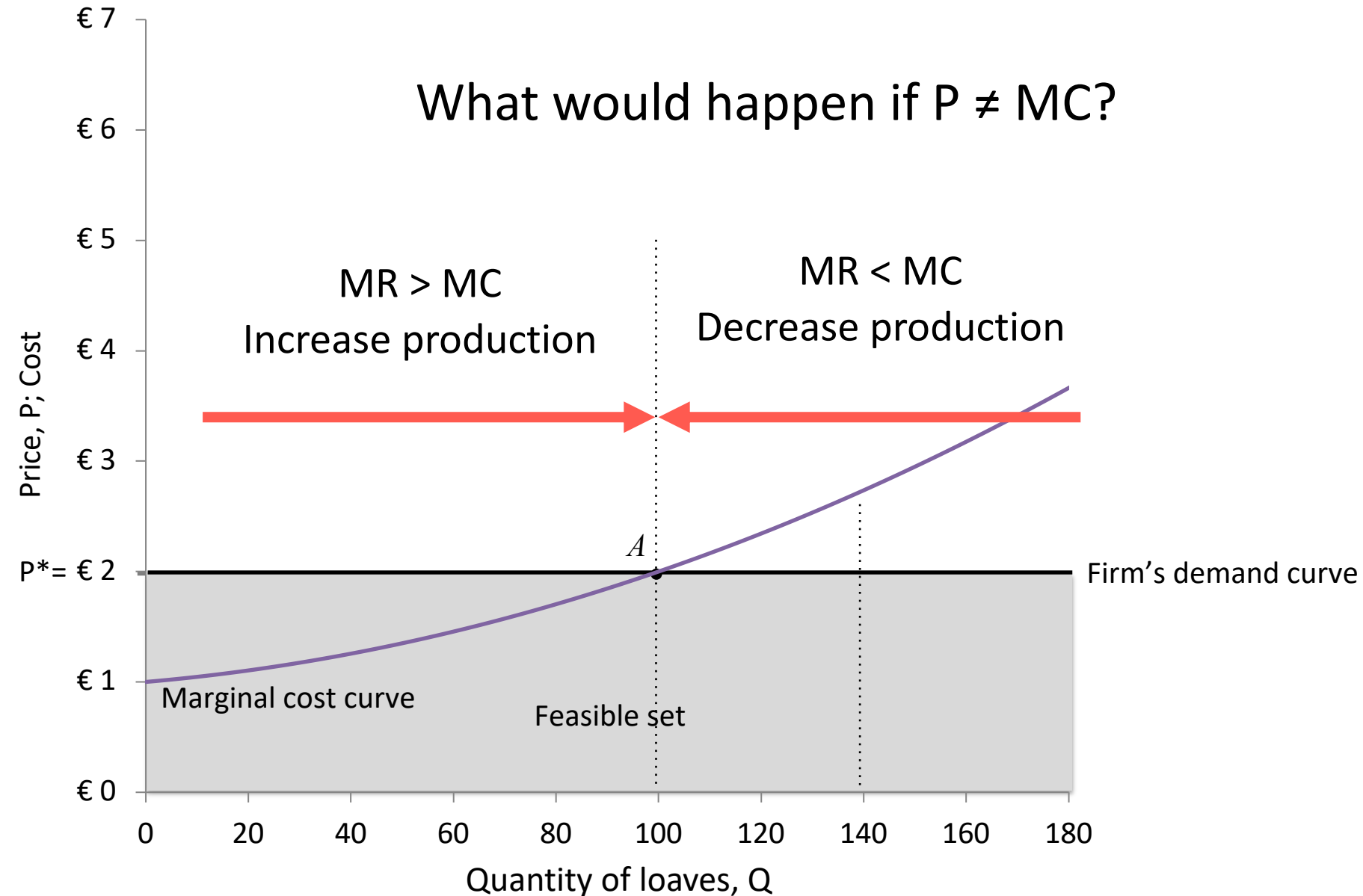


For price-taking firms, $MR = P$
(TR increase by P when Q increases by 1)
Maximize profit by choosing the quantity where $MR = MC$

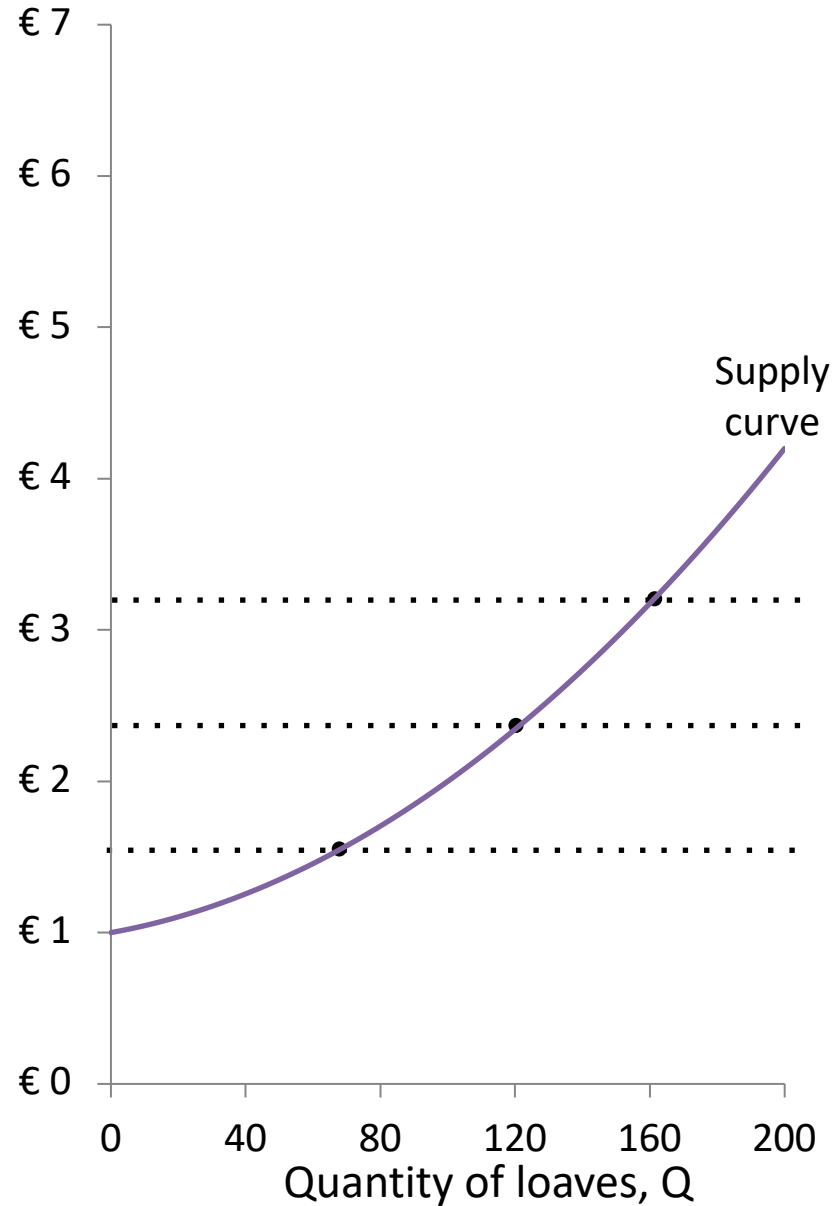
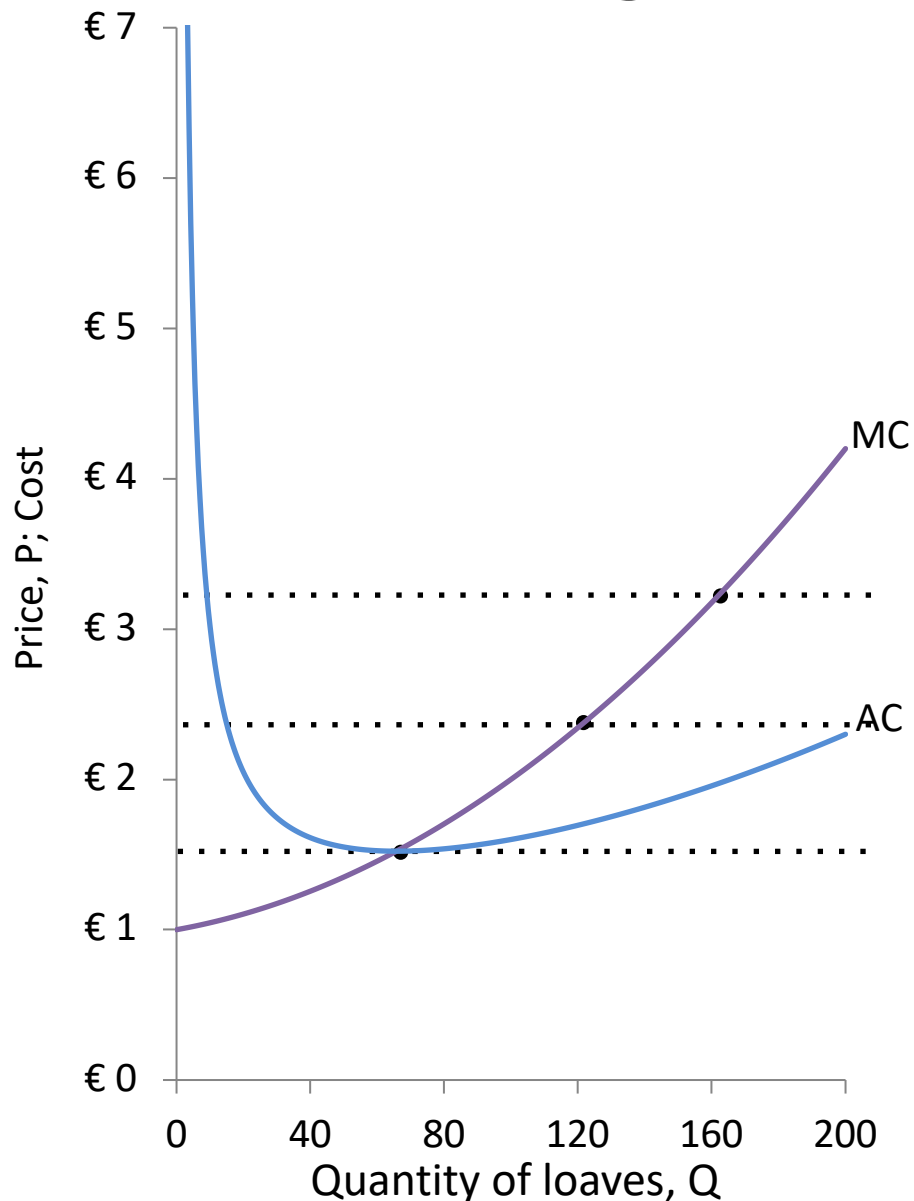
At **A**,
 $MR = P = MC$
Price-taking firms **choose**
the quantity where $P = MC$.

Price-taking firms

What would happen if $P \neq MC$?

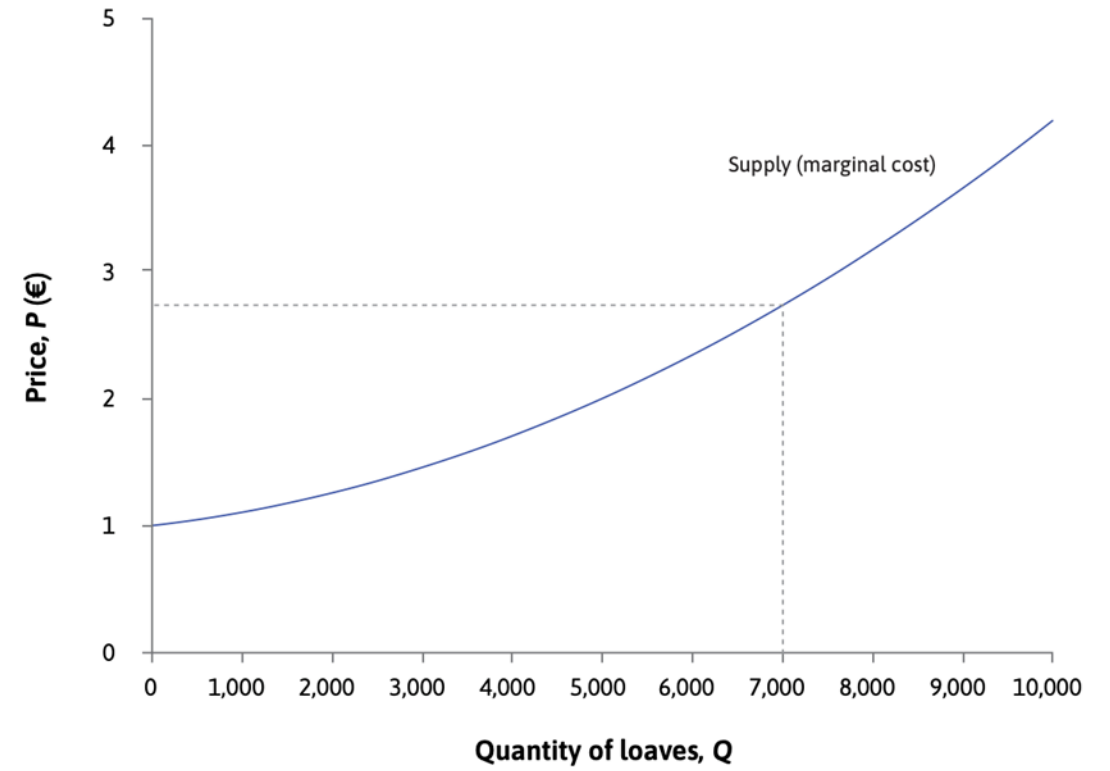
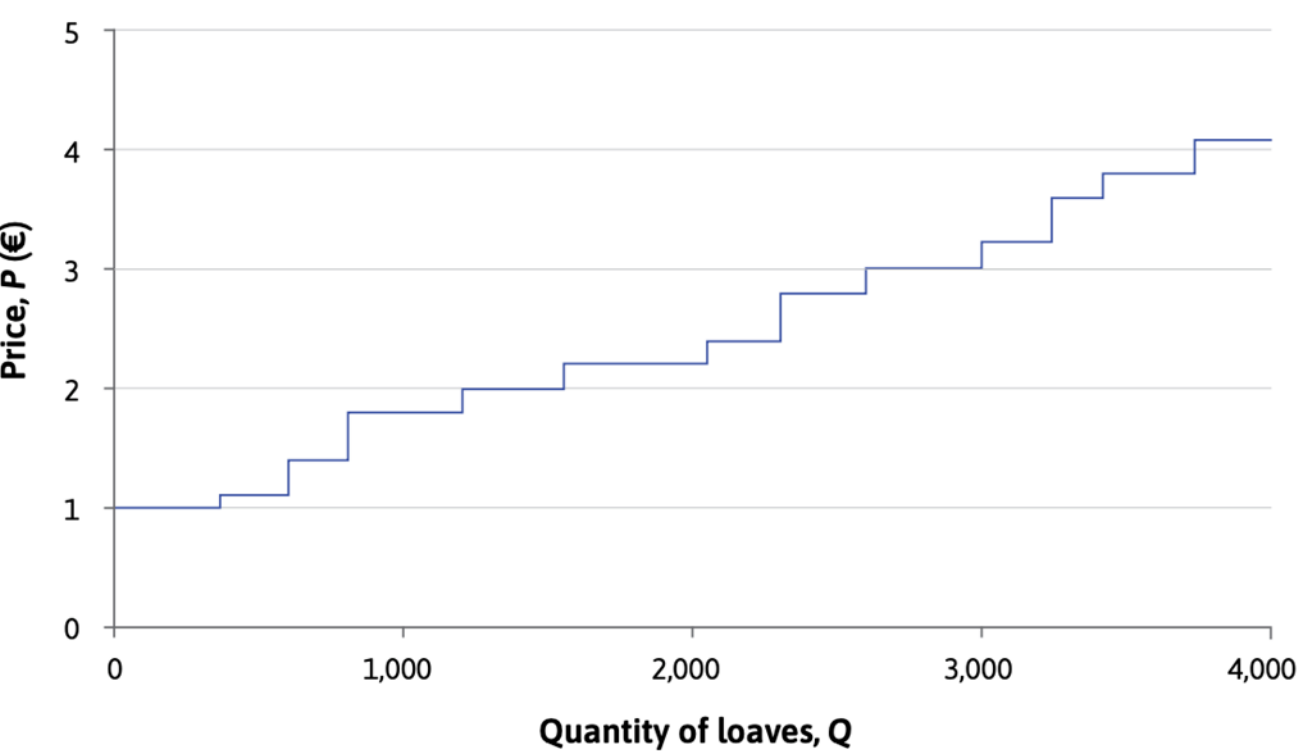


Price-taking firms

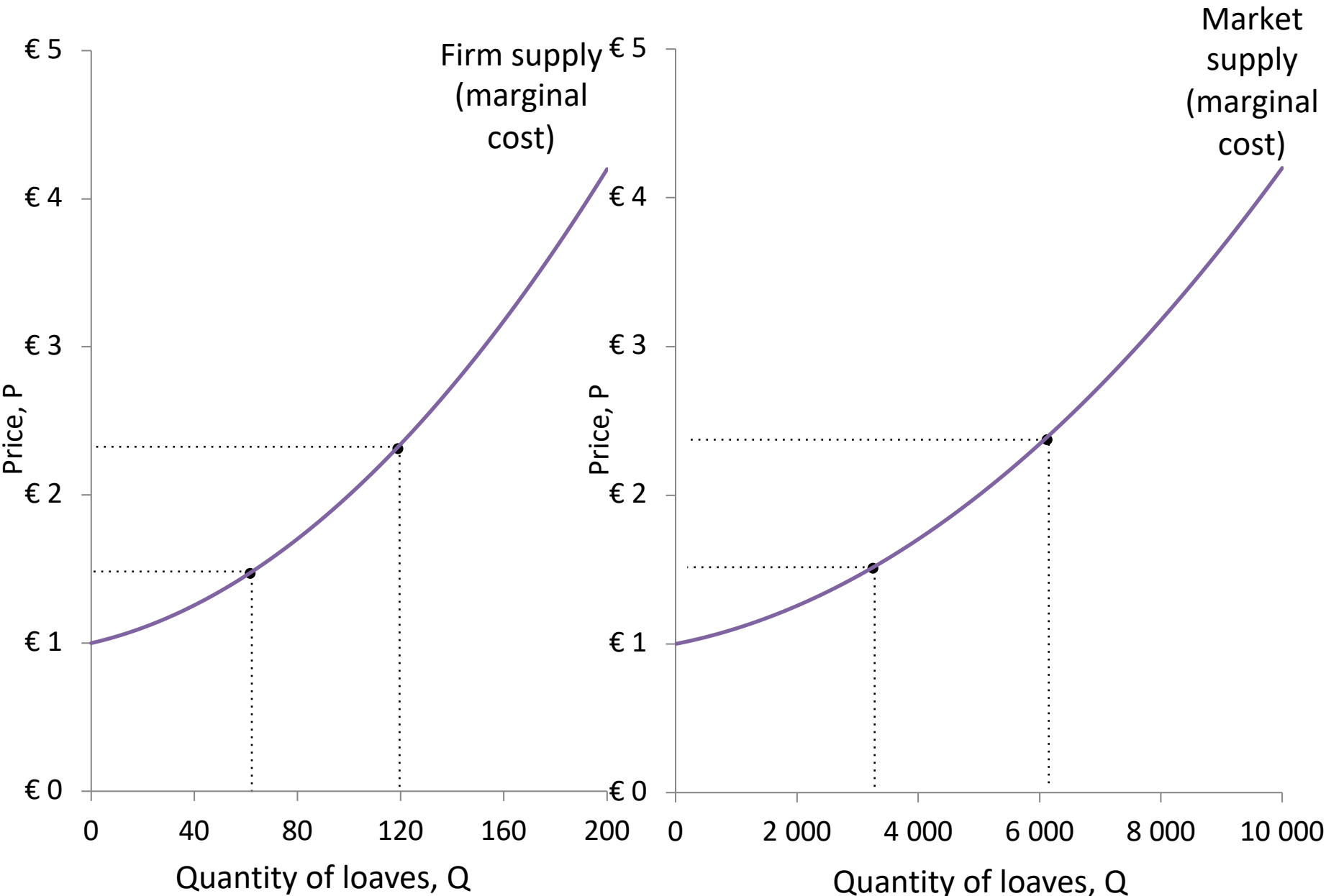


- Your supply depends on the market price:
 - €2.35 → 120 loaves
 - €3.20 → 163 loaves
 - €1.52 → 66 loaves
- If the price falls below €1.52:
 - $P < AC$
 - You make a loss in the short run
 - Decision to stay in business depends on expectations about future prices
- The **MC curve** is the **supply curve** for price-taking firms

Fig 8.9 & 8.10

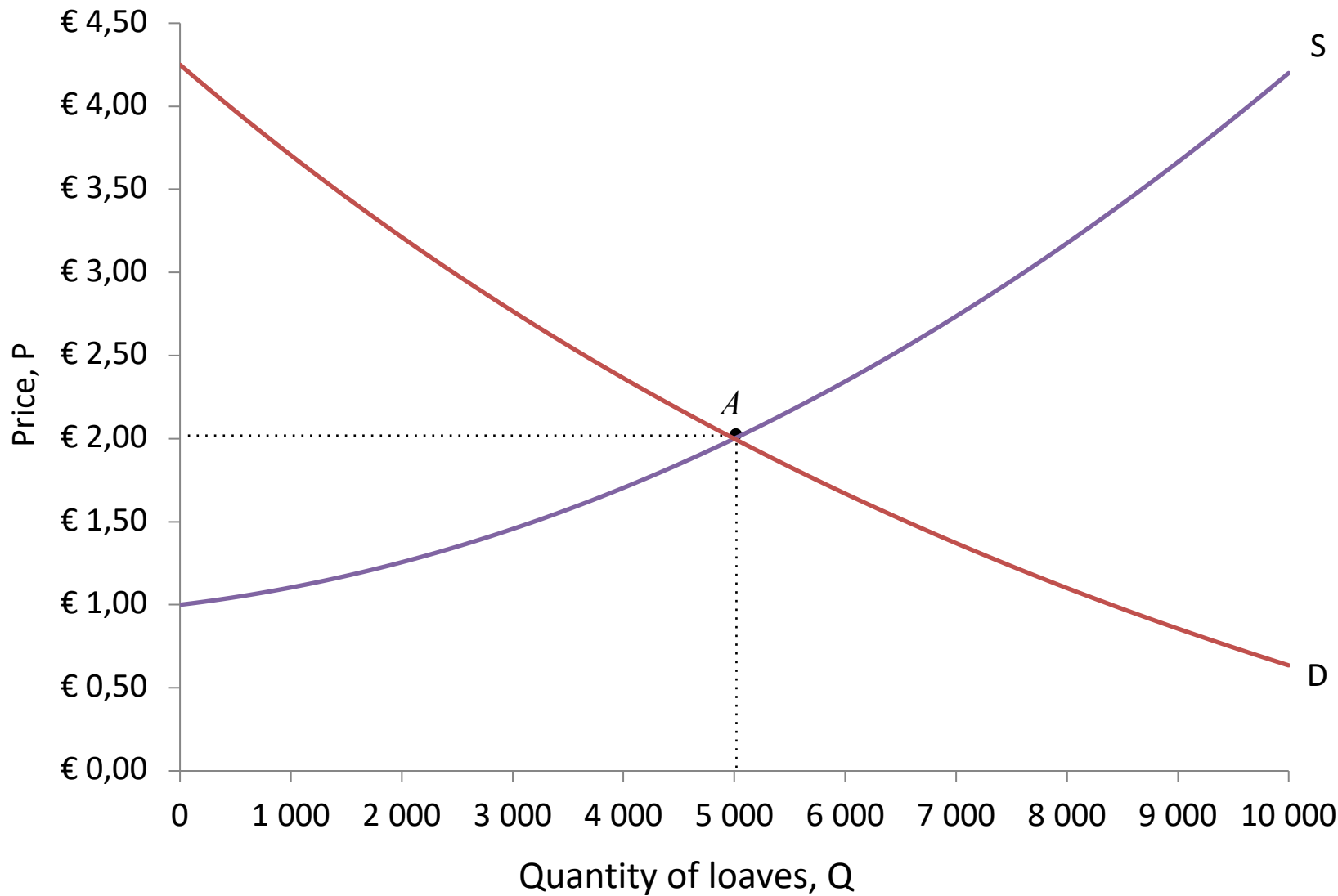


8.4 Market supply



- Suppose the city has 49 other bakeries with identical cost functions
- If the price is €2.35
 - Firm supplies 120
 - Market supplies $50 \times 120 = 6,000$
- If the price is €1.52
 - Firm supplies 66
 - Market supplies $50 \times 66 = 3,300$
- The **market supply** is the **horizontal summation of firm's supply curves**
- Market's supply curve is the market's MC curve
- The MC of the 6,001st loaf is about €2.35

Competitive market equilibrium



➤ We now have both the supply and demand curves



➤ The *competitive equilibrium* price is €2.00

- The market supplies 5,000.
- Each firm supplies $5,000/50 = 100$

➤ Does any individual or firm have an incentive to change behaviour?

- Would consumers purchase if $P > €2.00$
- Would firms produce if $MC < €2.00$

8.6 Changes in Supply and Demand



Quinoa is a traditional South American grain grown in the Altiplano.

Market for quinoa

- Quinoa recently became popular among consumers in the US and Europe.
- How has that impacted
 - the equilibrium price?
 - the equilibrium quantity?



Quinoa is the new oatmeal

Market for quinoa: data and theory

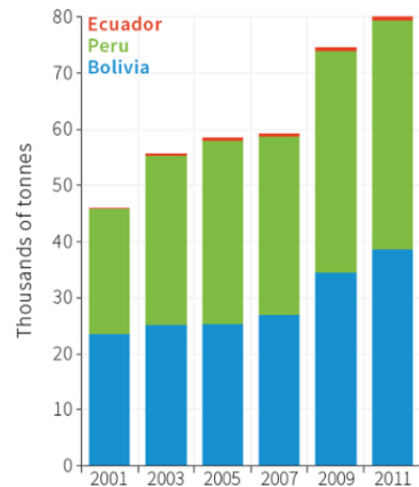


Figure 8.10a The production of quinoa. Source

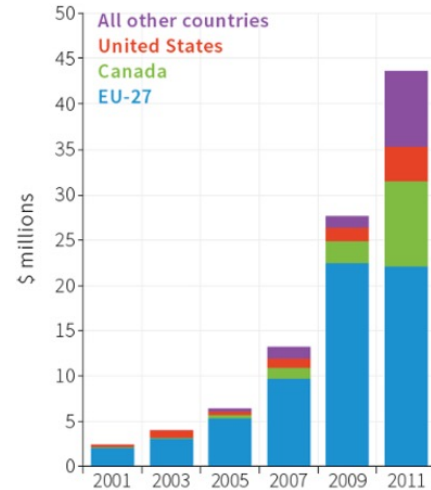


Figure 8.10c Global import demand for quinoa. Source

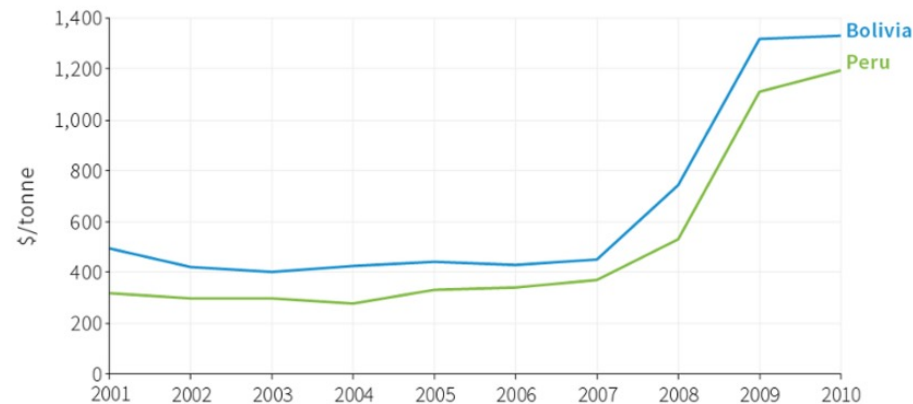
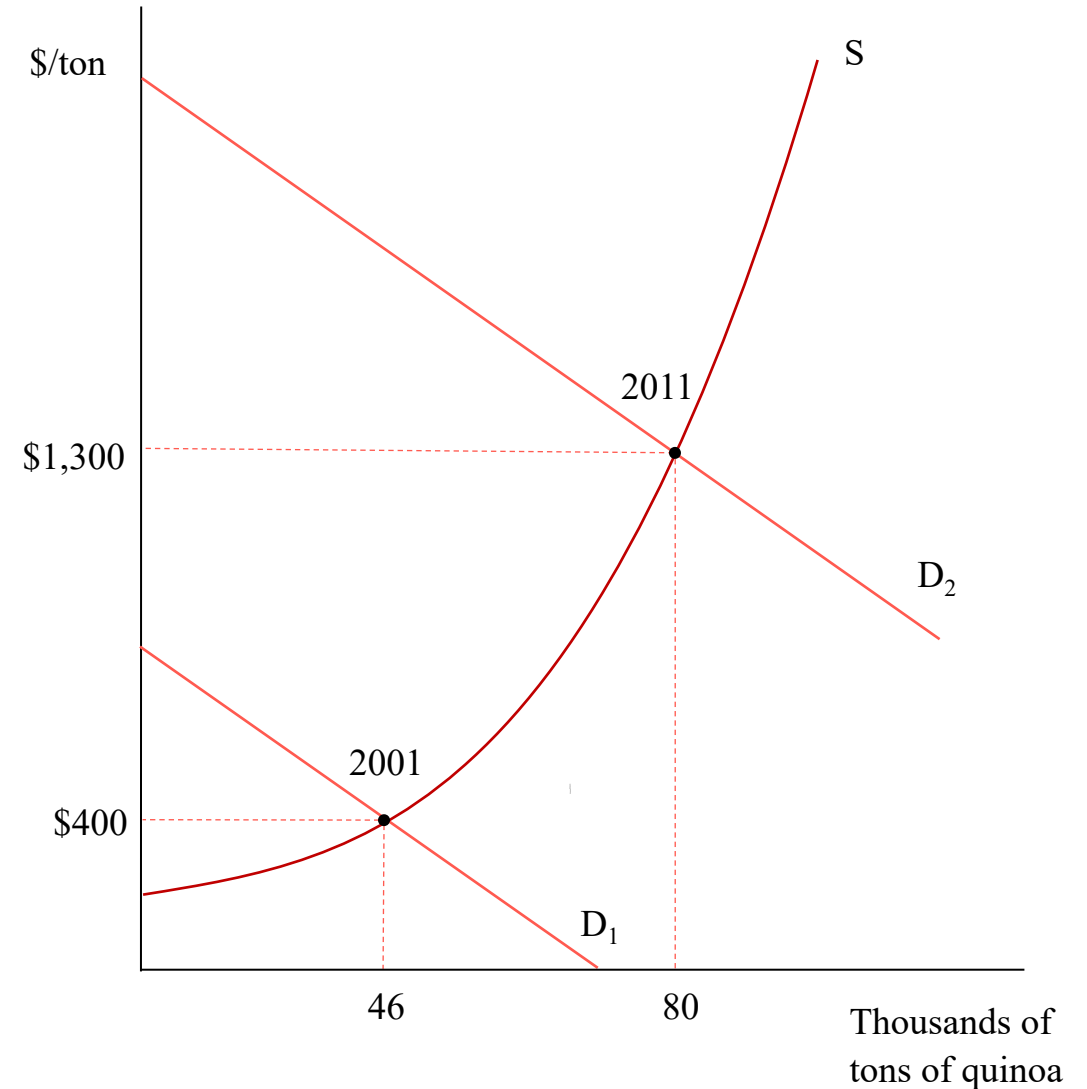
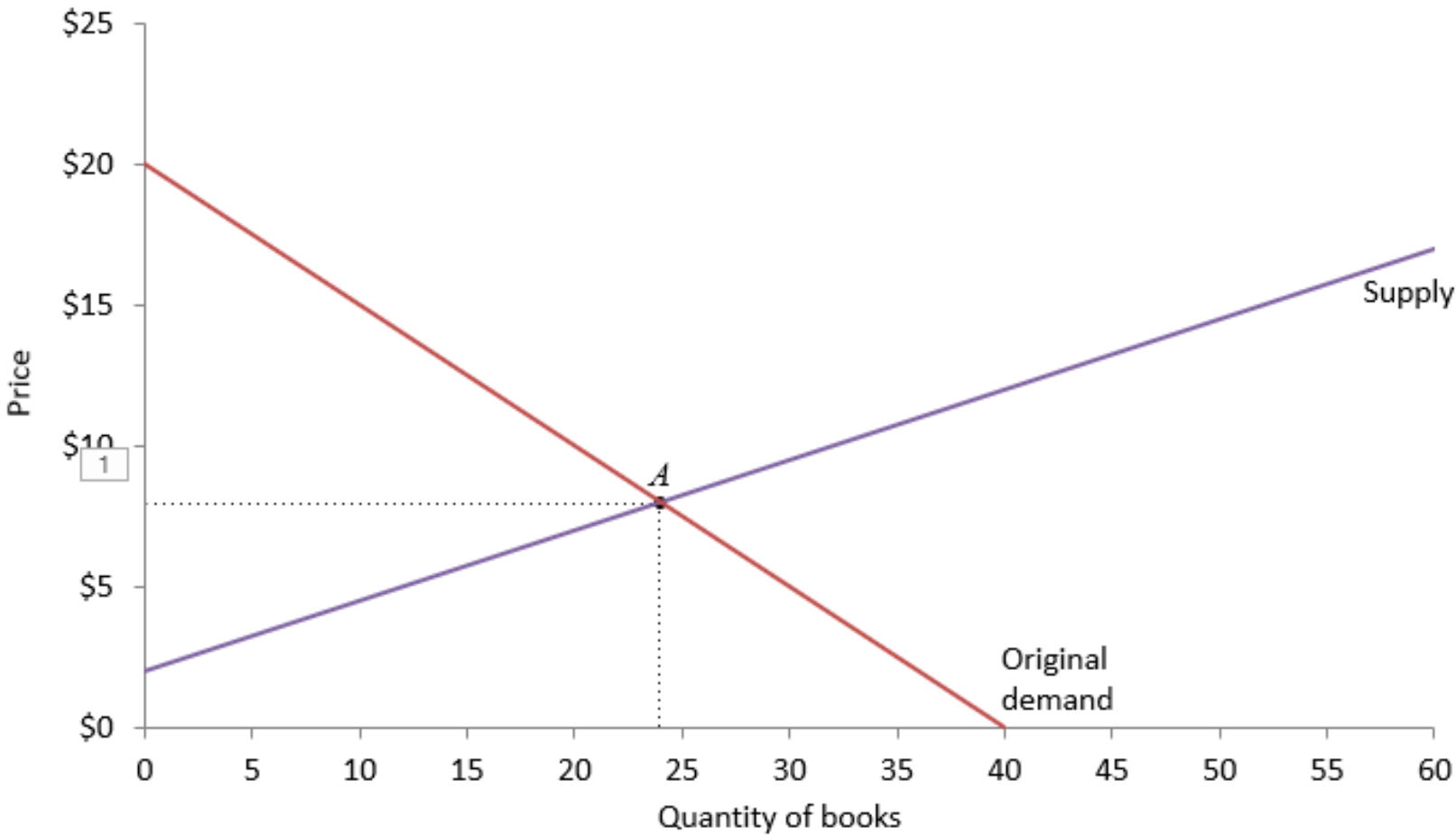


Figure 8.10b Quinoa producer prices. Source

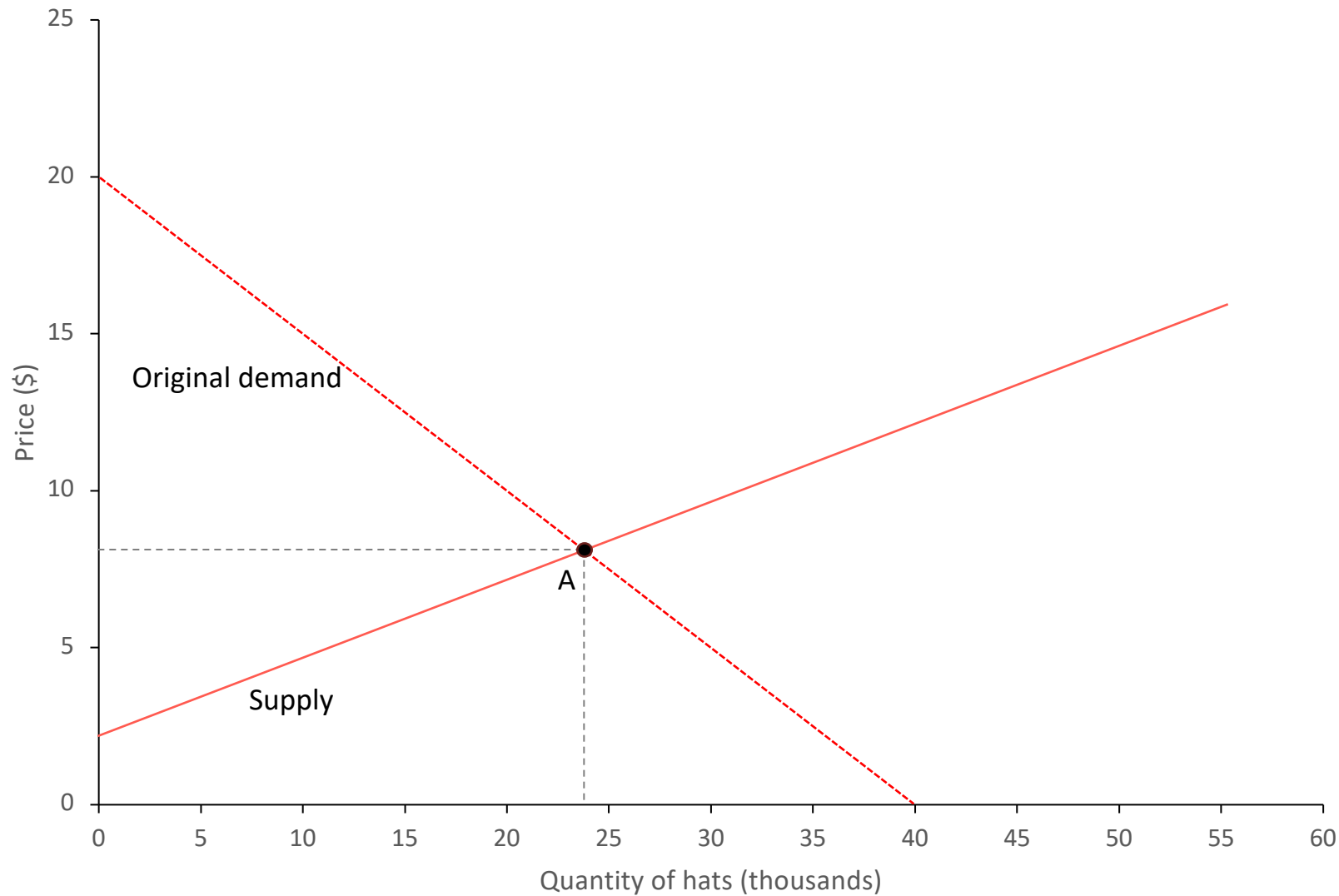


Changes in Supply and demand

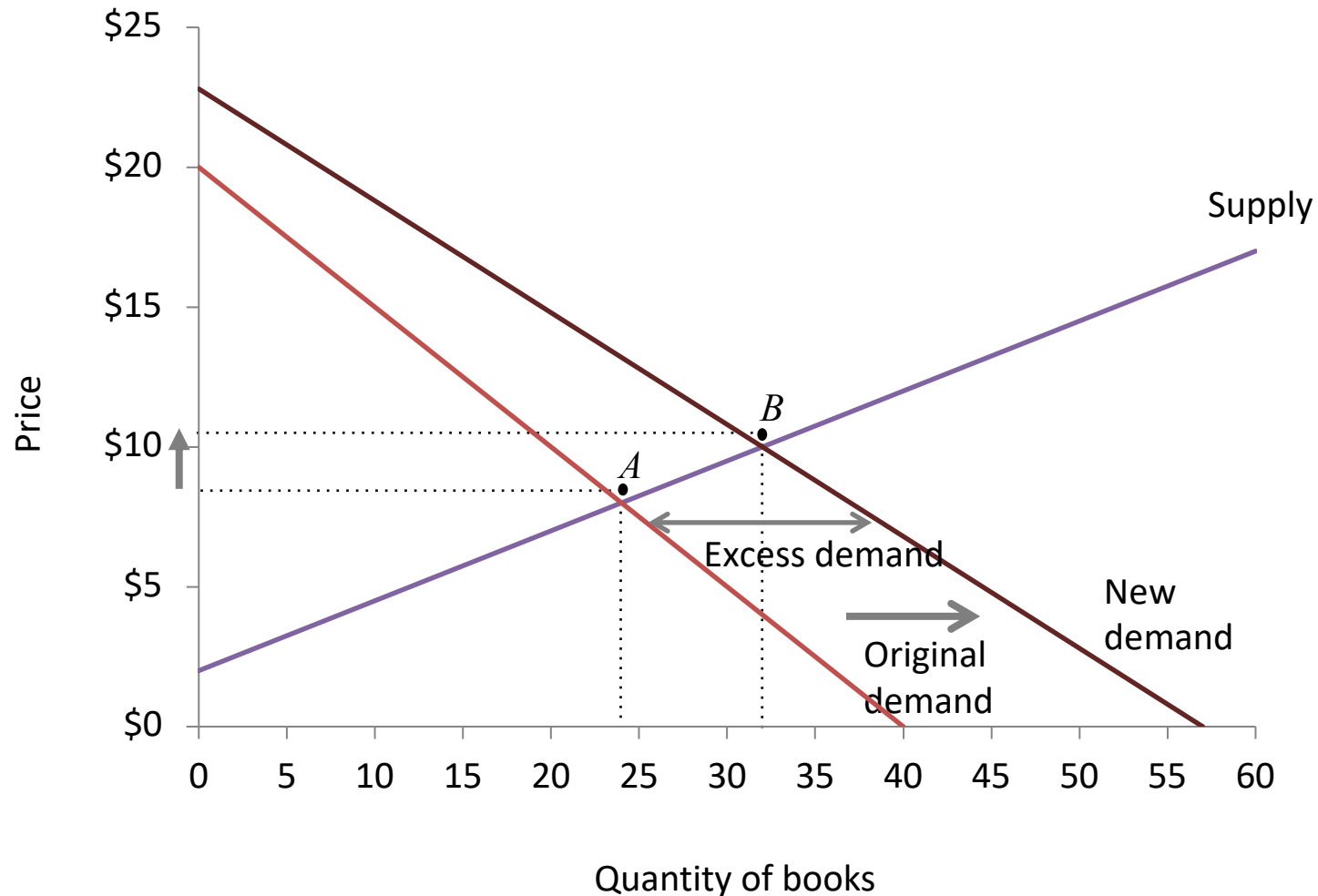


- Back to our market for secondhand textbooks
- Original competitive equilibrium is:
 - $Q^* = 24$ books
 - $P^* = \$8$.
- What happens to equilibrium prices when there's a change to either demand or supply?

Figure 8.14a – The initial equilibrium.

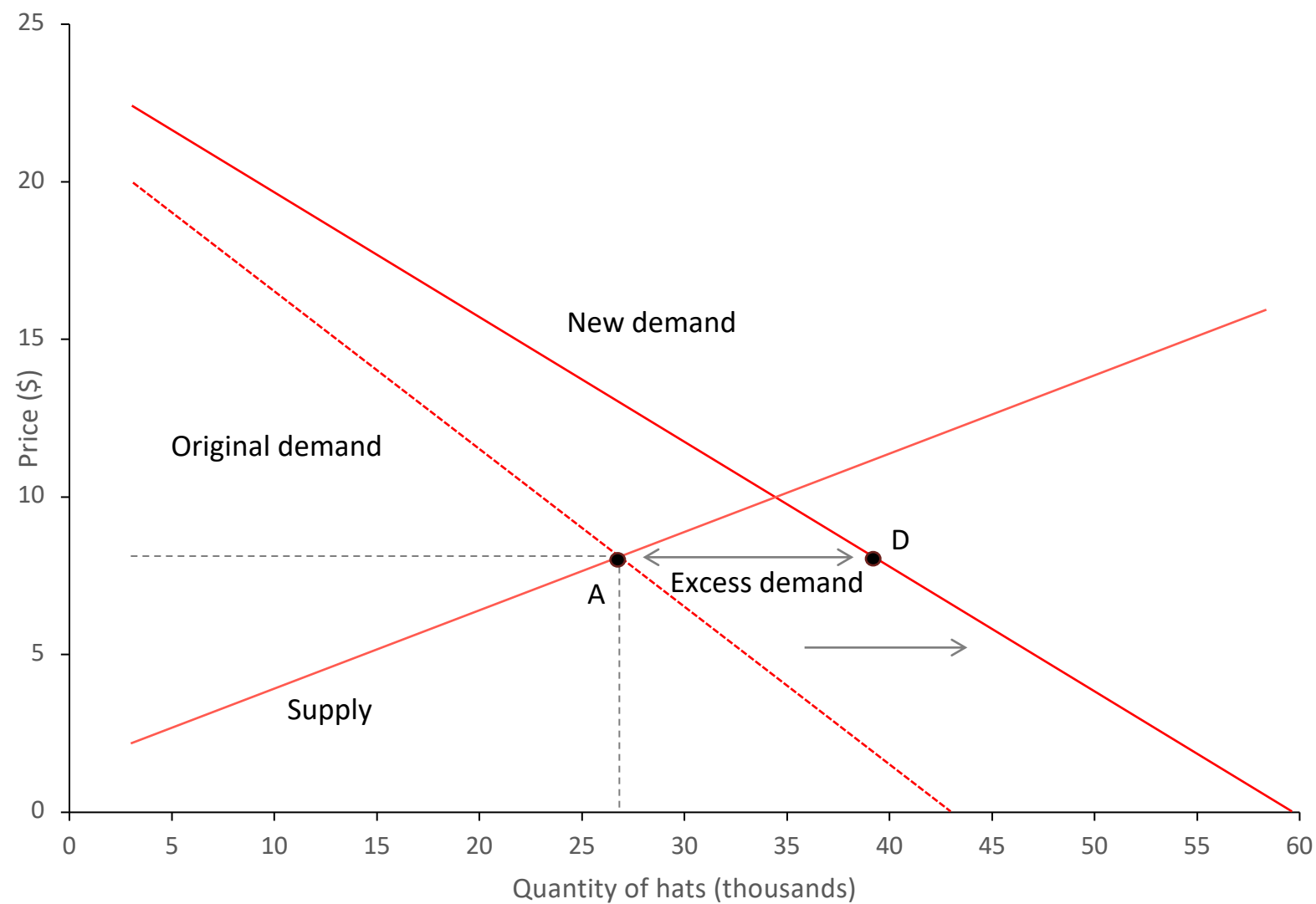


An increase in demand for textbooks

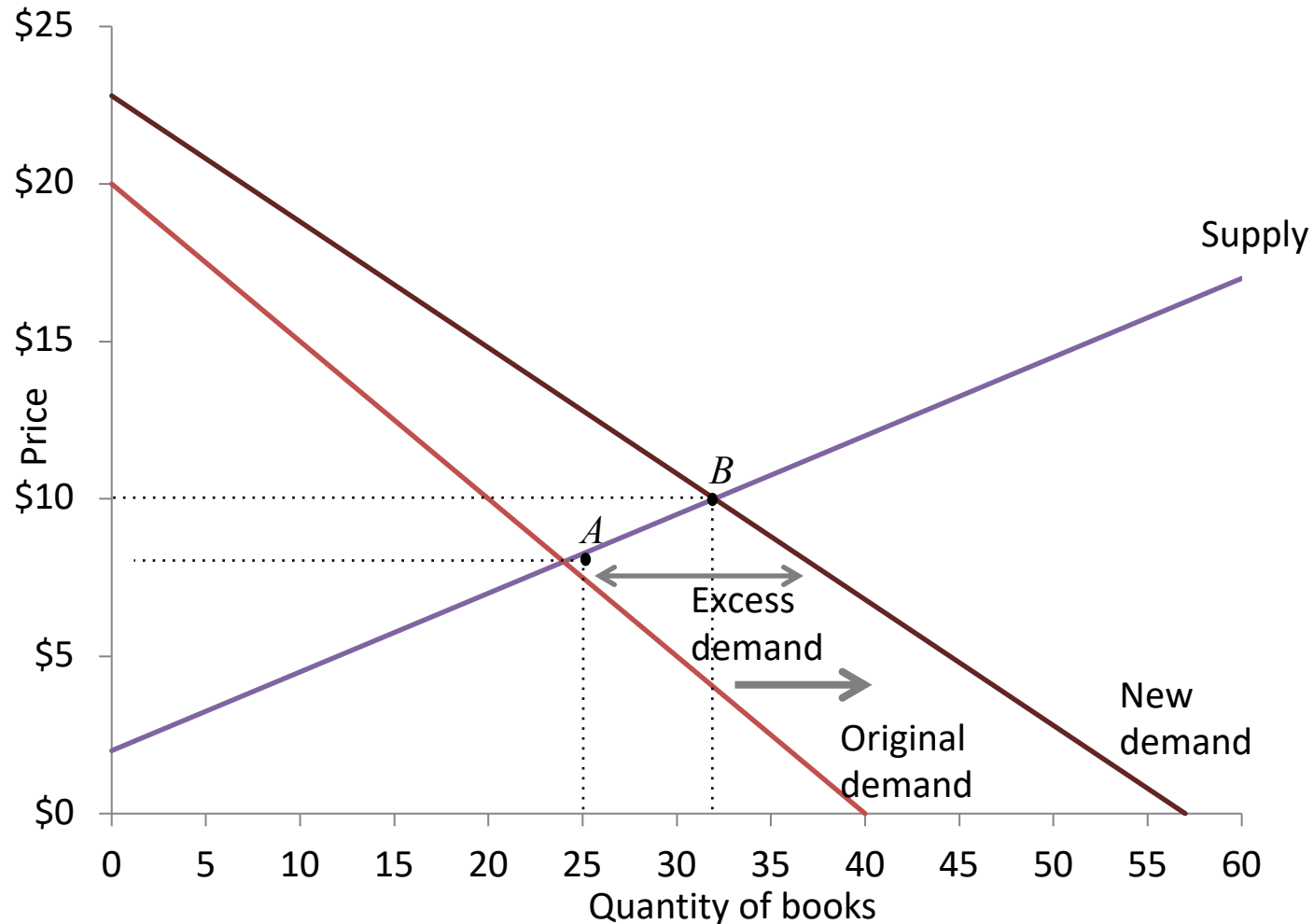


- Original competitive equilibrium is 24 books sold for \$8.
- Suppose course becomes more popular, there is *excess demand* at $P = \$8$
 - Demand curve shifts right
- Competition among buyers pushes the price up to *new equilibrium*
 - $P = \$10$ and $Q = 32$ books
- All the original sellers continue to sell their books, along with some new sellers
- Some of the original buyers no longer buy

Figure 8.14c – Excess demand.



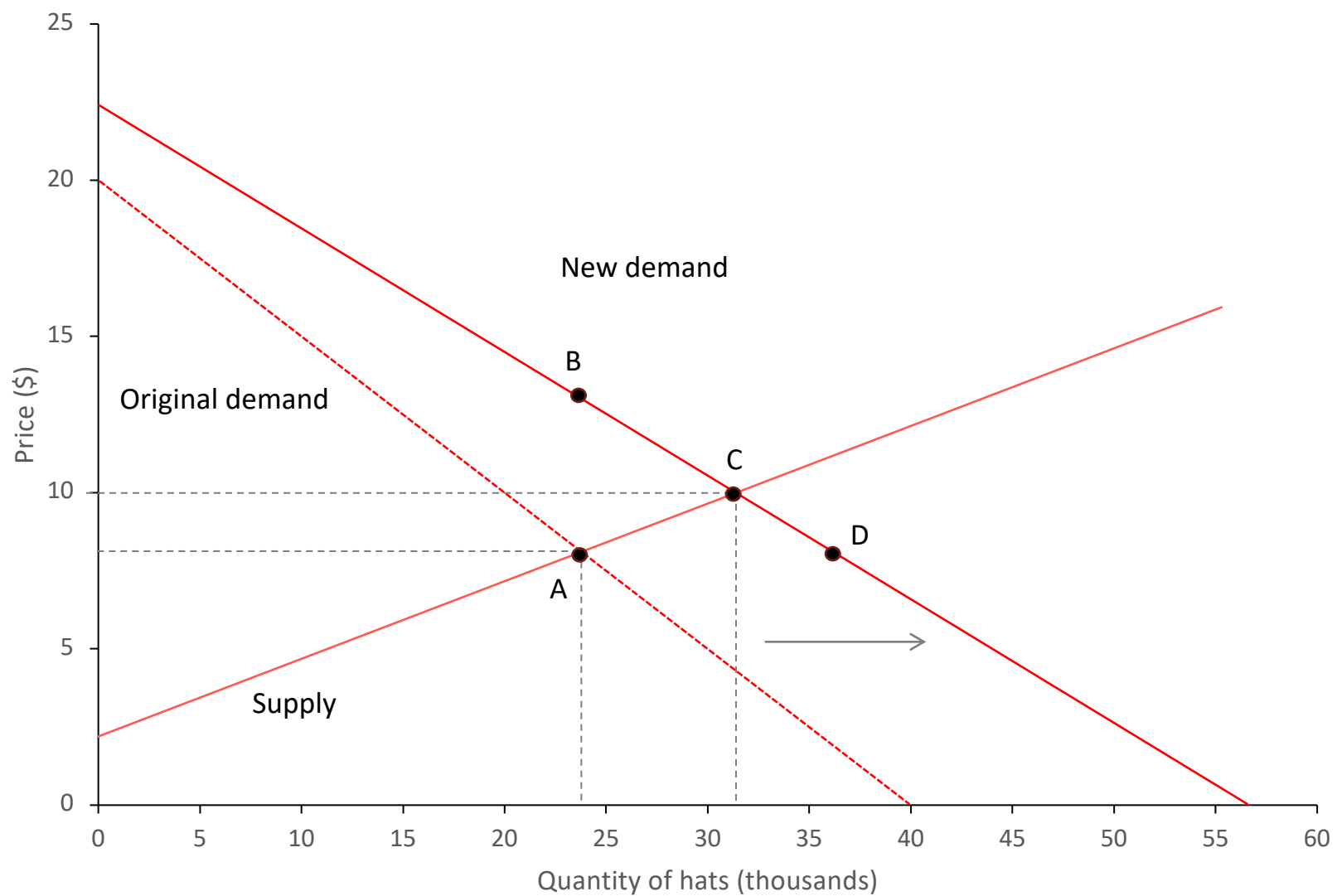
An increase in demand for textbooks



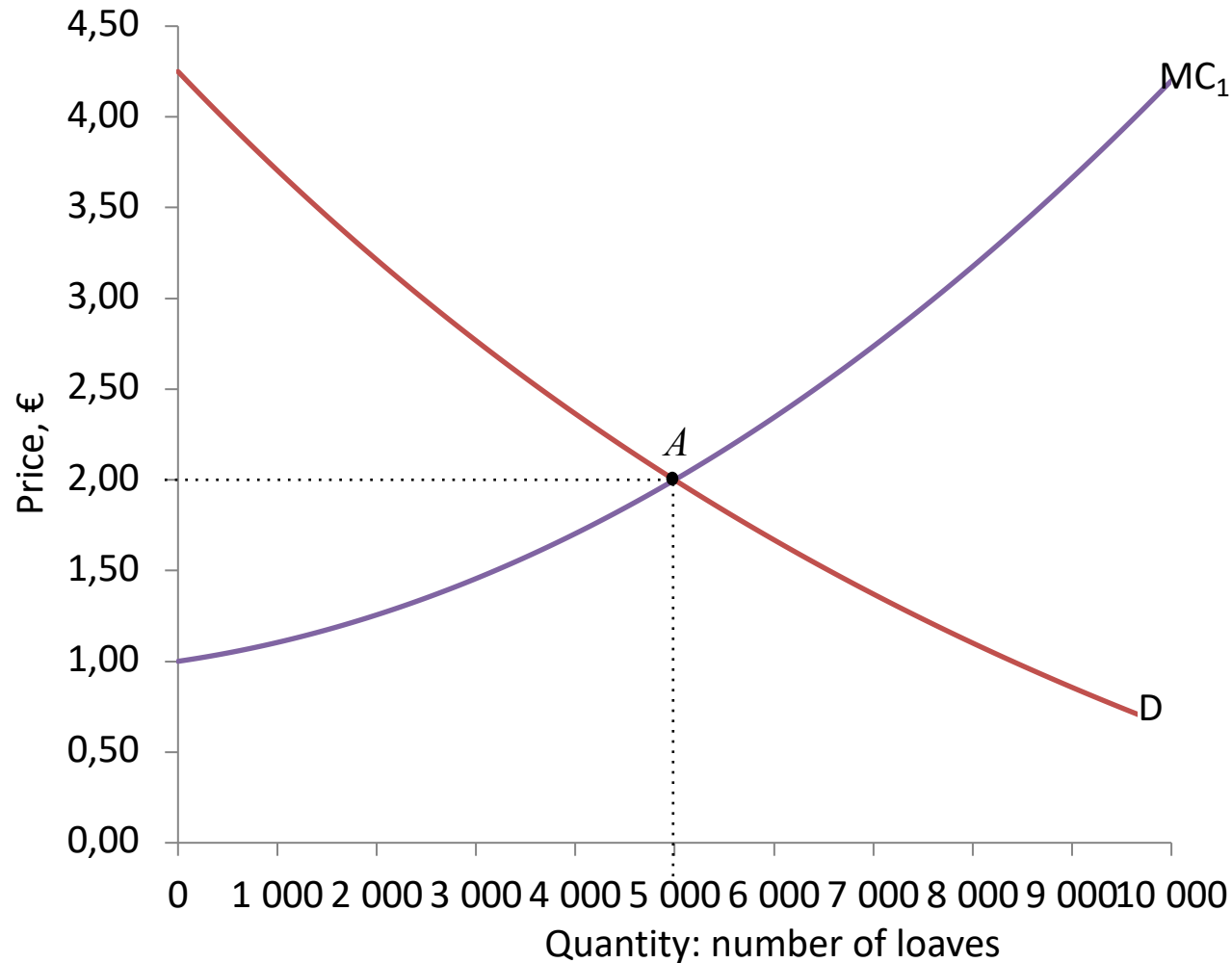
When the course becomes more popular:

- there is an “increase in demand”
 - the *quantity demanded* increases for any price
 - Shift of demand curve
- the equilibrium *quantity supplied* increases too
 - movement along the supply curve
 - the supply curve does not shift, and there is no “increase in supply”
 - Later, in Part 4, we will look at **how much** the price increases.

Figure 8.14d – Raising price and quantity.

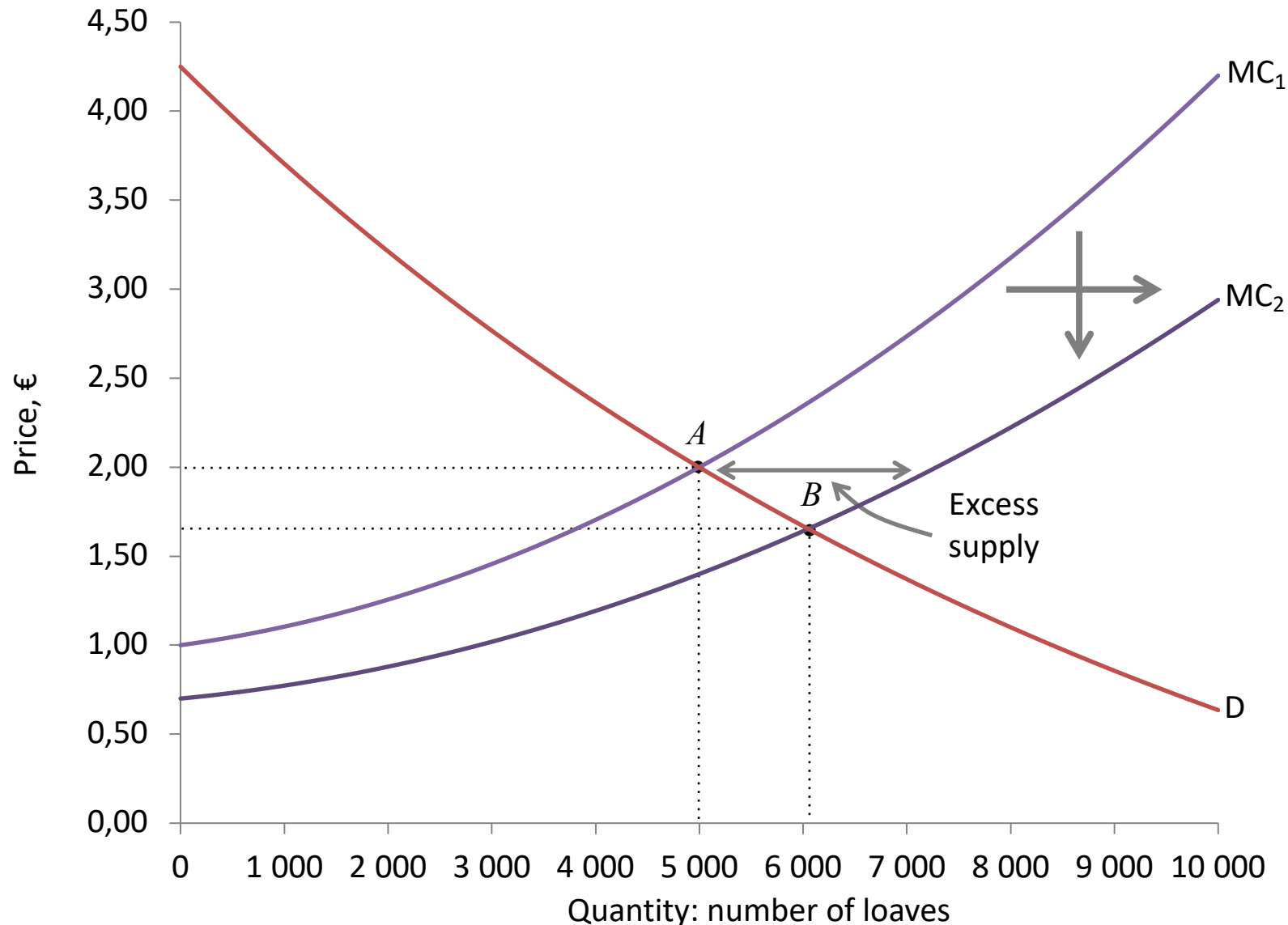


An increase in supply: *increase in productivity*



- Recall that the market supply curve is the marginal cost of bread in the *competitive market*
- Original equilibrium is
 $Q^* = 5,000$
 $P^* = €2$
- Now suppose the bakeries discover a technique that reduces the cost of baking bread

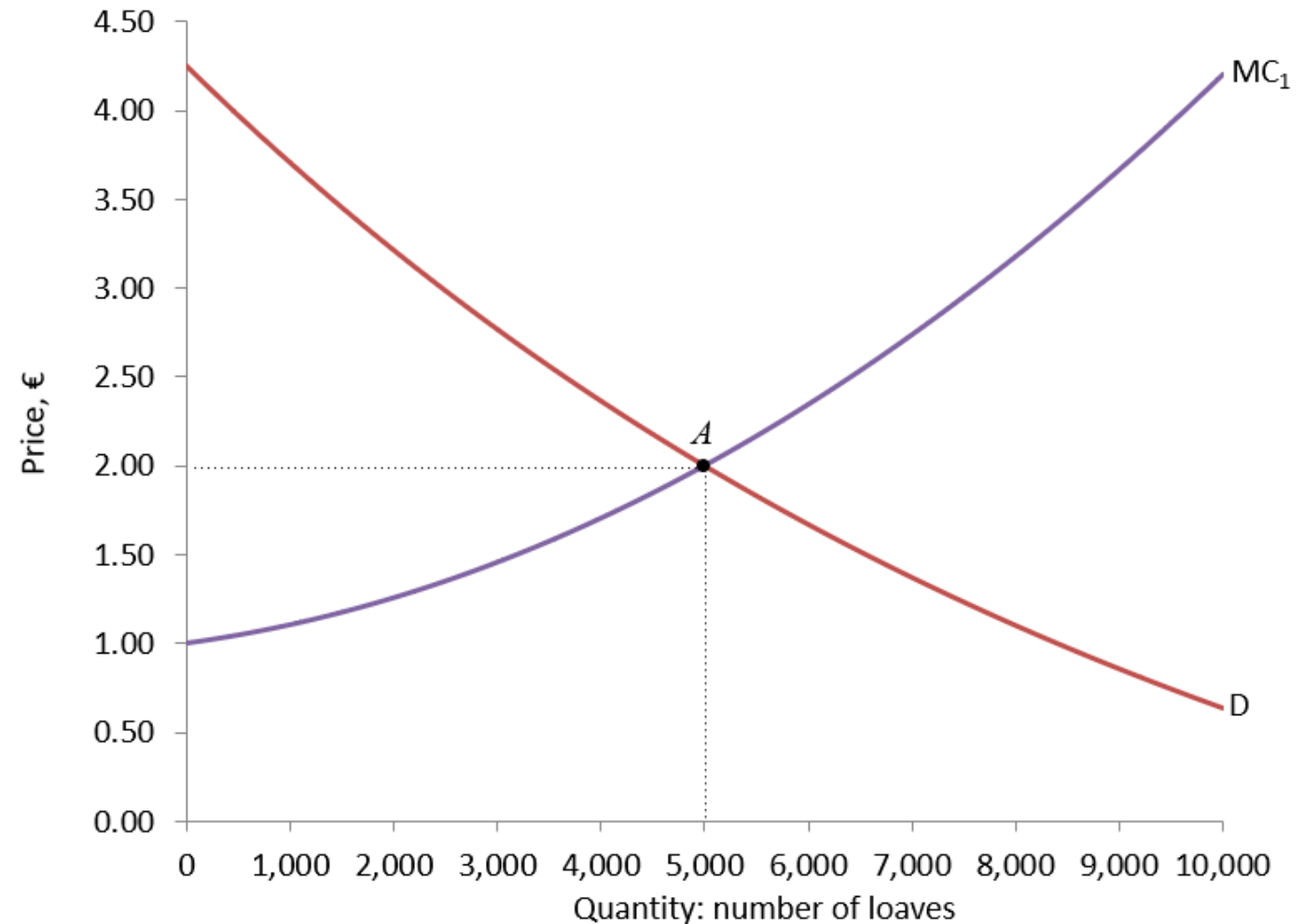
An increase in supply: *increase in productivity*



- Original equilibrium at A
 $Q^* = 5,000$
 $P^* = €2$
- New technique that reduces the cost of baking bread
 - This lowers the MC curve
 - Or shifts out the supply curve
- Now there is *excess supply* at $P = €2$.
- Competition among buyers pushes the price down until new equilibrium at B
 $Q = 6,000$
 $P = €1.70$

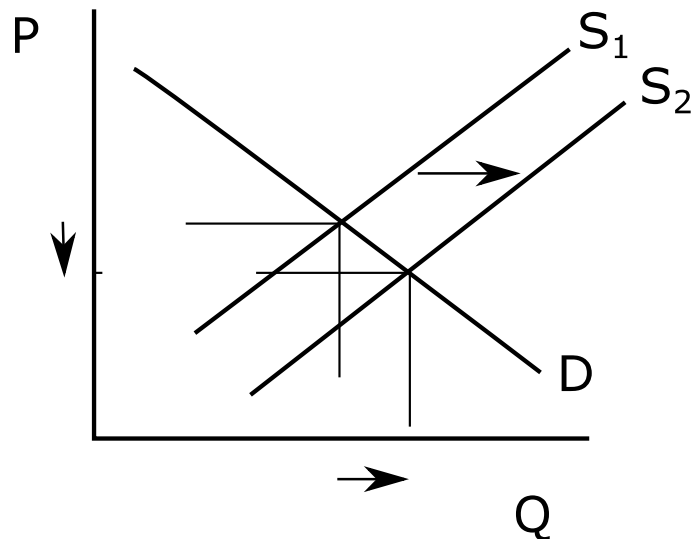
Changing market equilibria

- Changes in consumers' *WTP* shifts the demand curve
- Changes in firms' costs shift the supply curve
- What would happen to the short-run equilibrium price and quantity if:
 - the price of wheat increases?
 - consumers adopt low-carb diets?
 - ovens become more energy efficient?
 - the price of butter increases?
 - a decline in pizza sales pushes some pizzerias into the bread market?
 - unemployment falls?

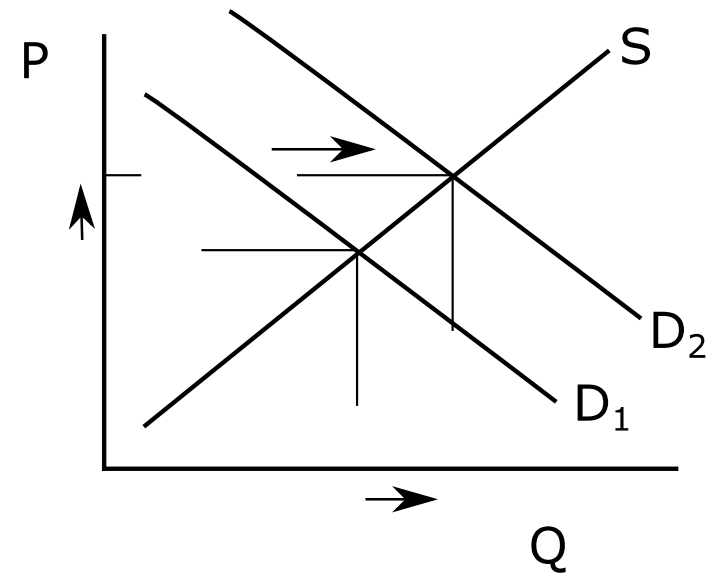


Changing market equilibria

- Supply increases when
 - Price of inputs decrease
 - Technology reduces MC
 - Number of firms in market increase
 - State of nature, e.g. a good harvest



- Demand increases when
 - Preferences change
 - Price of a complement falls
 - Price of a substitute increases
 - Average income and/or number of buyers increase



An increase in supply: *increase in number of bakeries*

➤ Suppose we now have an increase in the number of bakeries

- This also increases the supply of bread

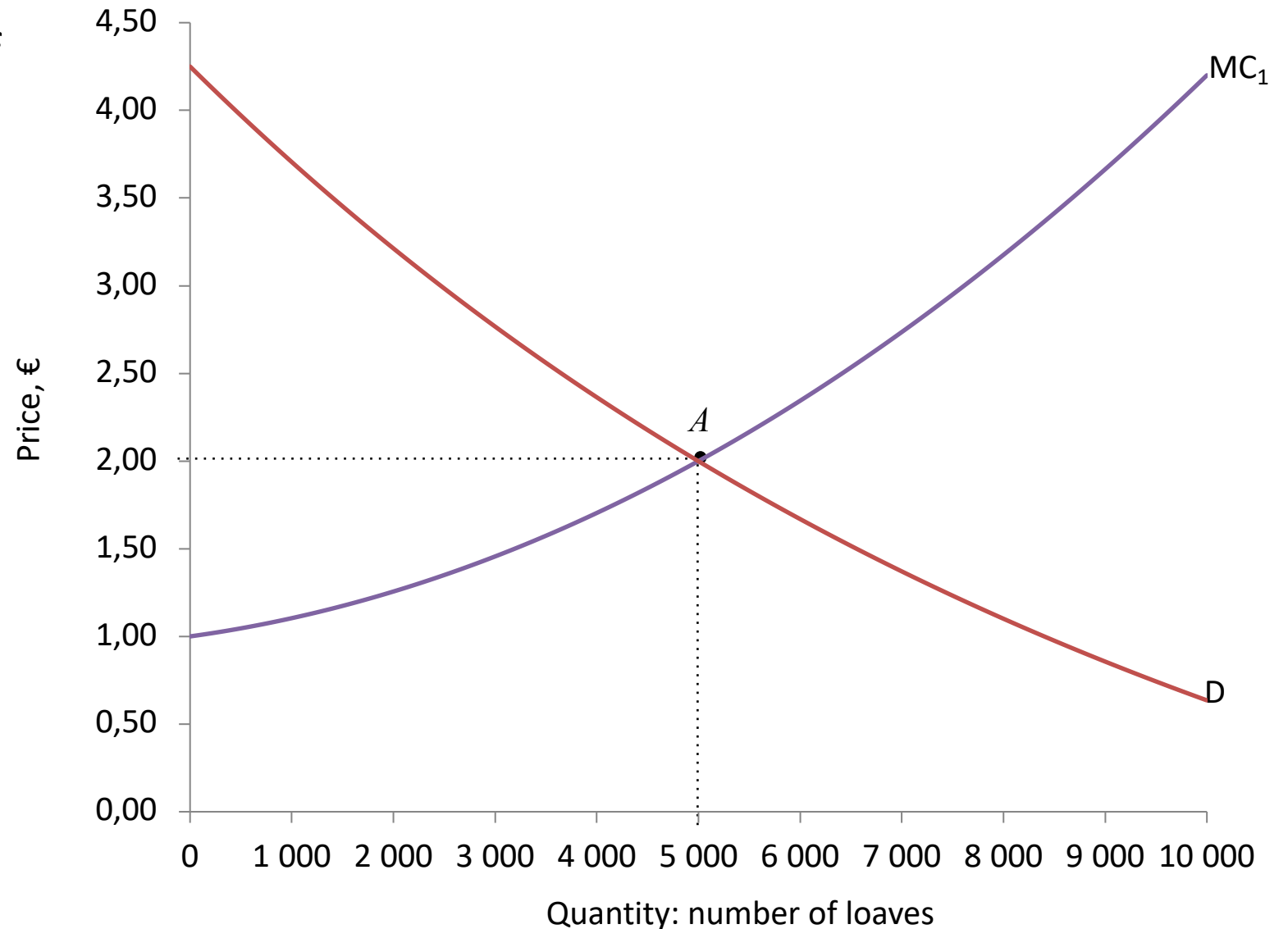
➤ Original equilibrium:

$$P^* = €2$$

$$Q^* = 5,000$$

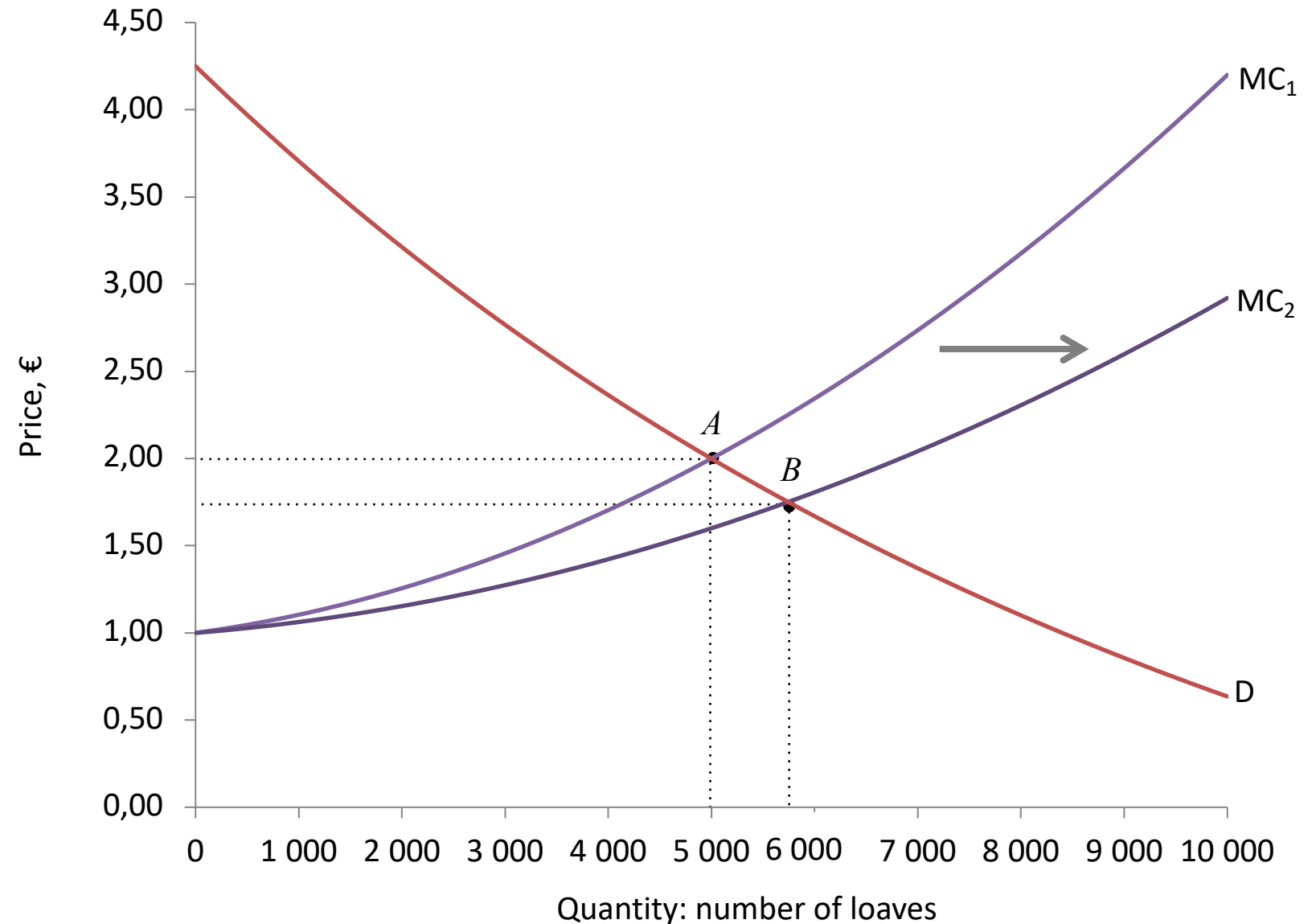
- 50 bakeries produce 100 loaves each.
- All firms earn a profit because $P > AC$

➤ If profits are positive, and the costs of entry are small, more firms will enter the market

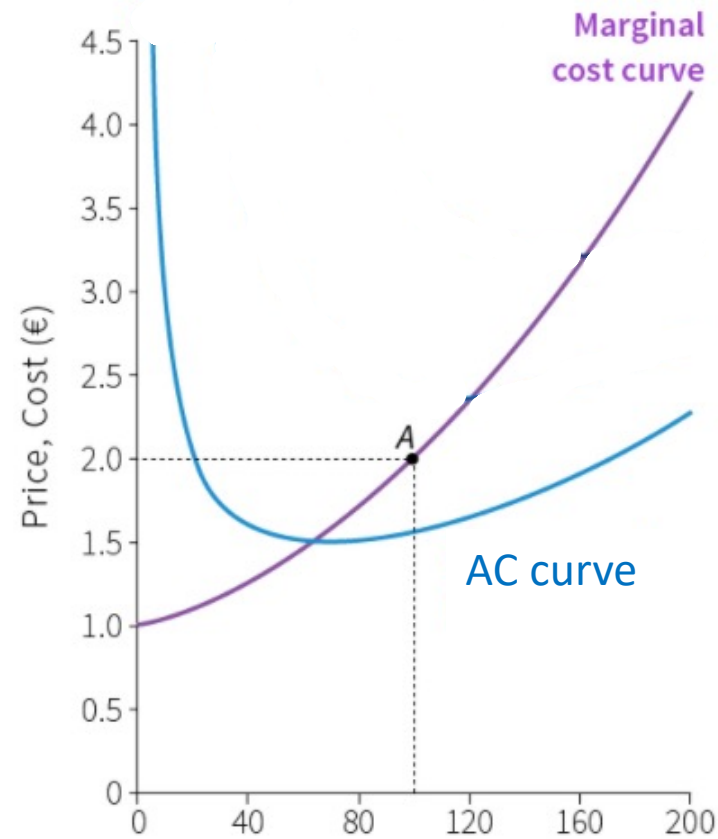


An increase in supply: *increase in number of bakeries*

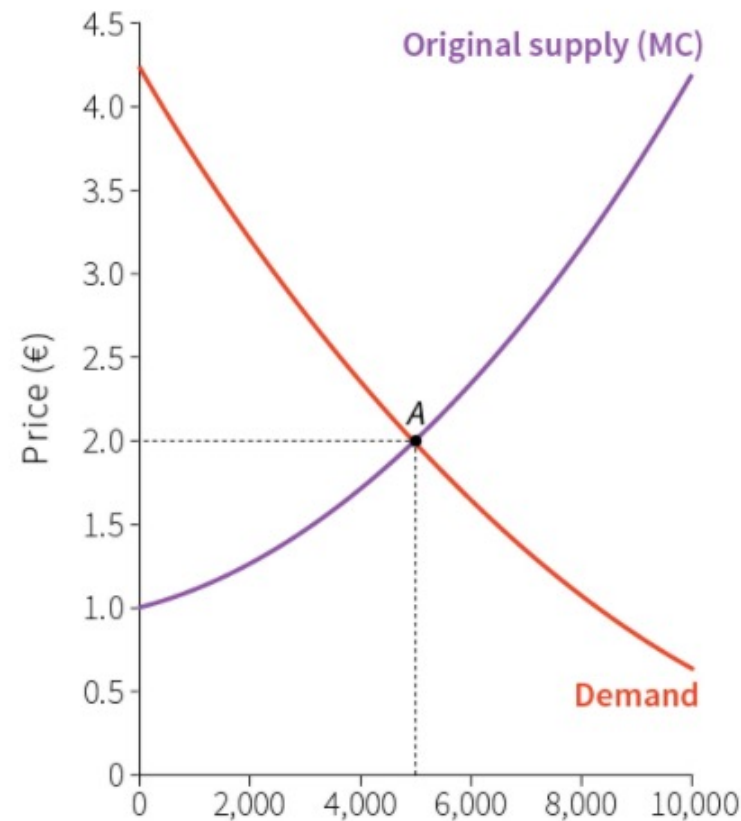
- **Economic profit incentive**
- When firms enter the market, the market supply curve shifts out
- Remember, the market supply simply adds up the quantity supplied by every firm at each price
- An increase in the number of bakeries will reduce the equilibrium price and increase the equilibrium quantity.
- How long will firms enter this market?



An increase in the number of bakeries

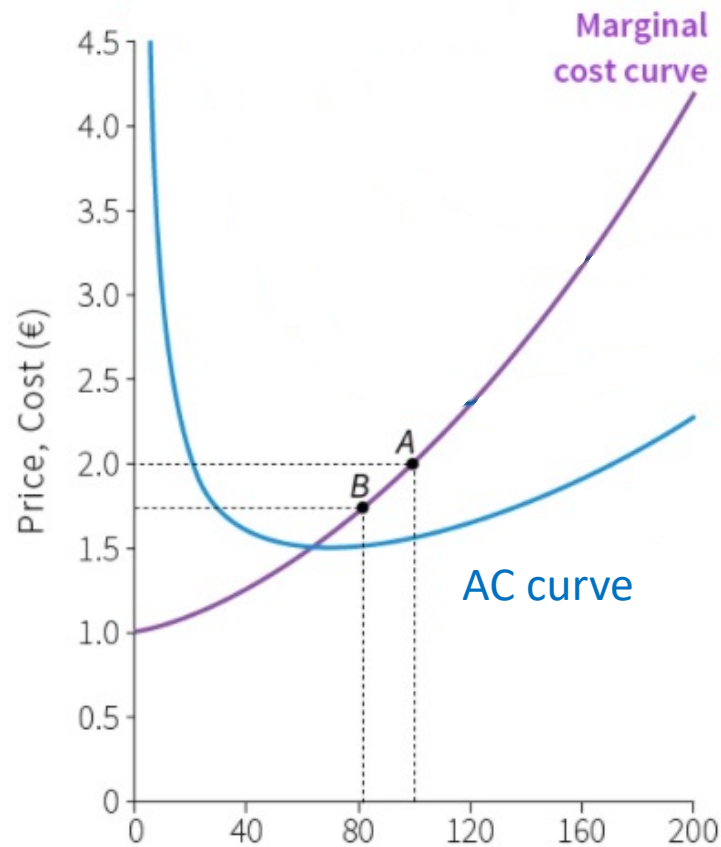


Quantity Q: number of loaves

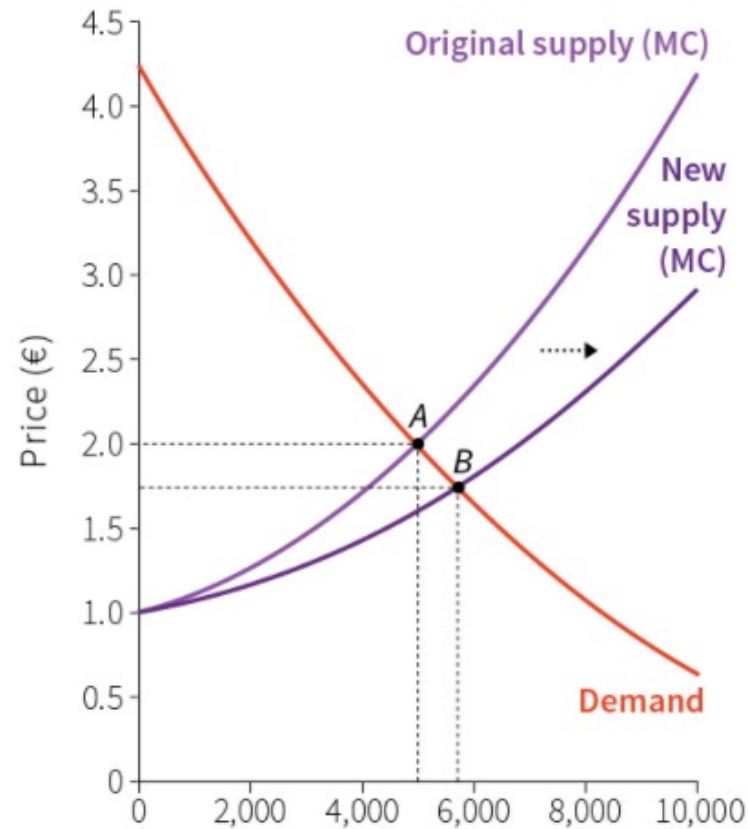


- Firms will enter the market as long as they can earn a positive profit
- At A, firms earn a profit

An increase in the number of bakeries



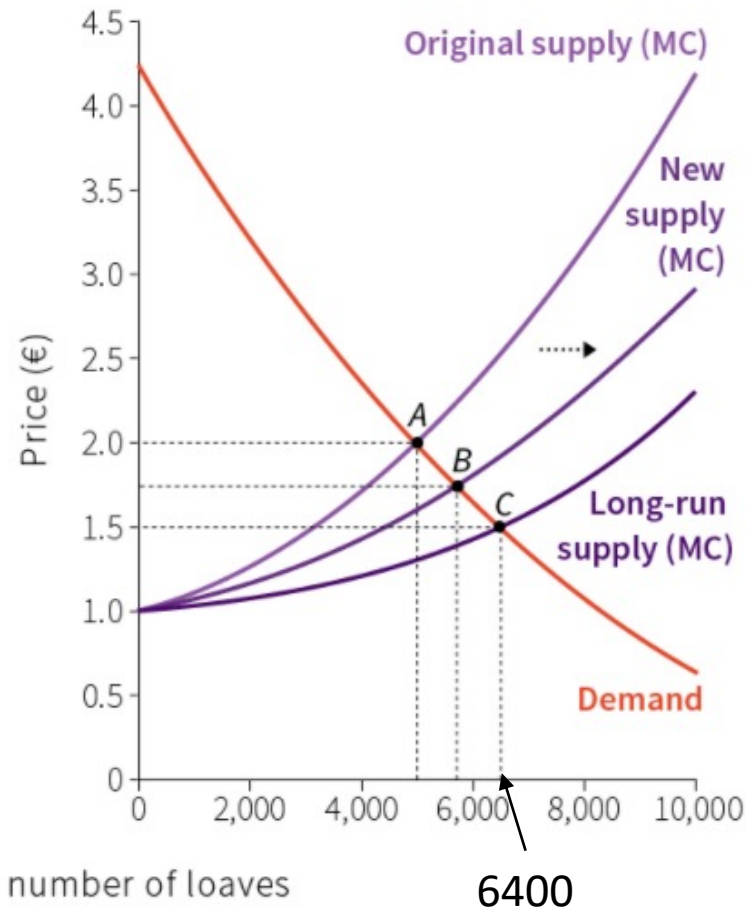
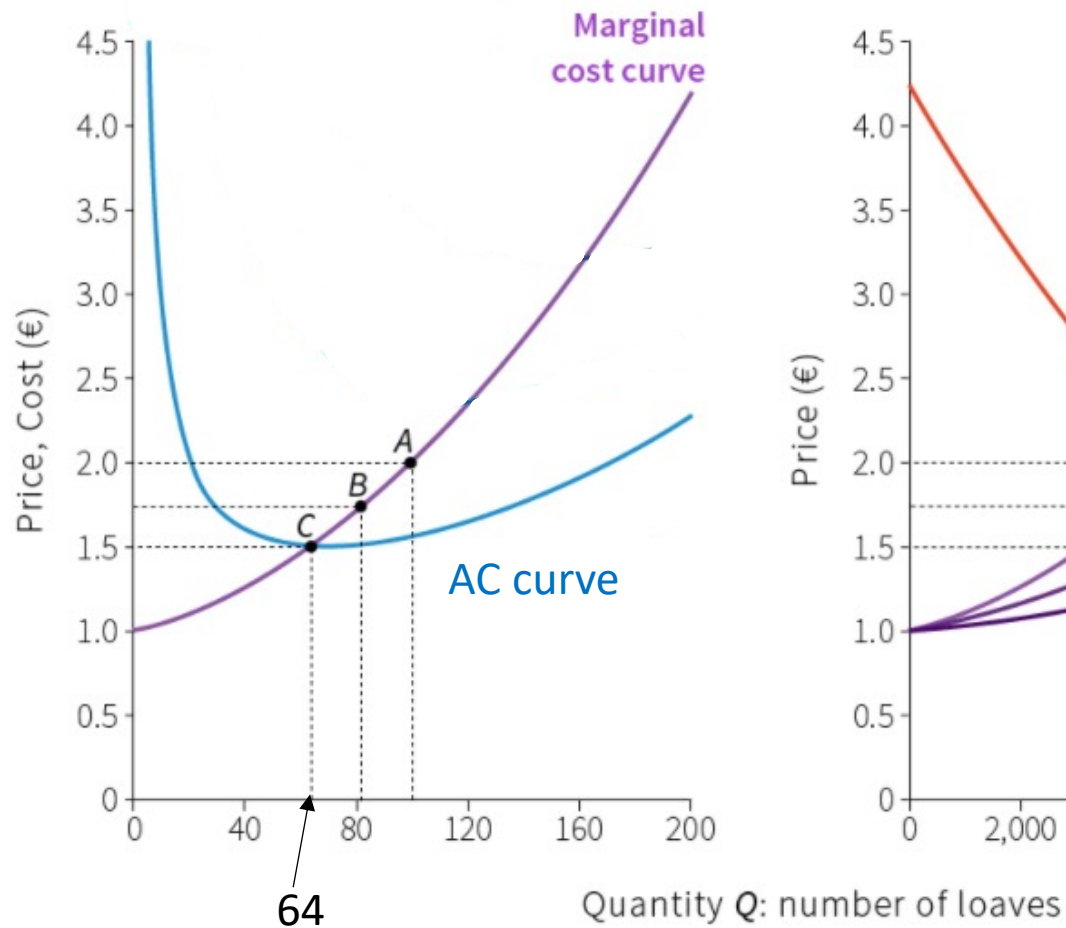
Quantity Q: number of loaves



➤ Firms will enter the market as long as they can earn a positive profit

- At A, firms earn a profit
- At B, firms earn a profit

An increase in the number of bakeries



➤ Firms will enter the market as long as they can earn a positive profit

- At A, firms earn a profit
- At B, firms earn a profit
- At C, firms earn 0 profit

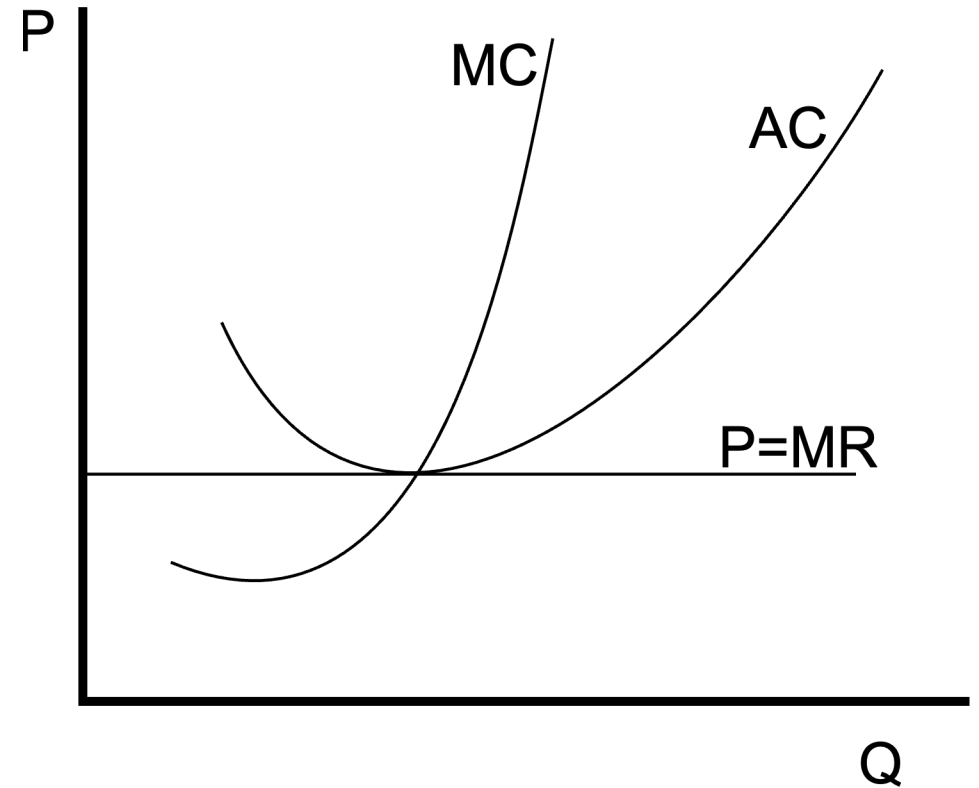
➤ In the long run, the competitive equilibrium is:

$$\begin{aligned}P &= \text{€}1.50 \\Q &= 6,400. \\ \pi &= 0\end{aligned}$$

- Each firm produces a quantity of 64
- There are 100 firms

Long-run competitive equilibrium

- Firms are maximising profit: $MR = P = MC$
- Zero economic profit: $P = AC$
- Therefore, $P = MC = AC$
- When $MC = AC$, AC is at its minimum
- Therefore, in LR equilibrium, price competition and entry / exit have driven price down to the minimum (long-run) average cost.
- The number of firms adjusts (in response to profit / loss incentives) to ensure the lowest possible price.
(assuming entry is cheap and easy)



Short-run and long-run equilibrium

Short-run equilibrium

- $Q_d = Q_s$
- $P = MR = MC$
- $P < AC$, $P = AC$ or $P > AC$
- Firms can make economic losses, break even or make economic profit
- To reach equilibrium: price adjusts, and individual firms adjust quantities produced

Long-run equilibrium

- $Q_d = Q_s$
- $P = MR = MC = AC$
- Economic profit = zero
- To reach equilibrium: price adjusts, individual firms adjust quantities and entry/exit takes place

Exercise

Suppose that firms selling bottled water in South Africa (a competitive market) are currently in long term equilibrium. Then, a natural disaster destroys a significant number of bottled water producers' factories.

1. Explain, using graphs, how each firm in the industry (the surviving ones) and the market as a whole adjust to a new short-run equilibrium.
2. Explain how the market and firms adjust to a new long-run equilibrium.



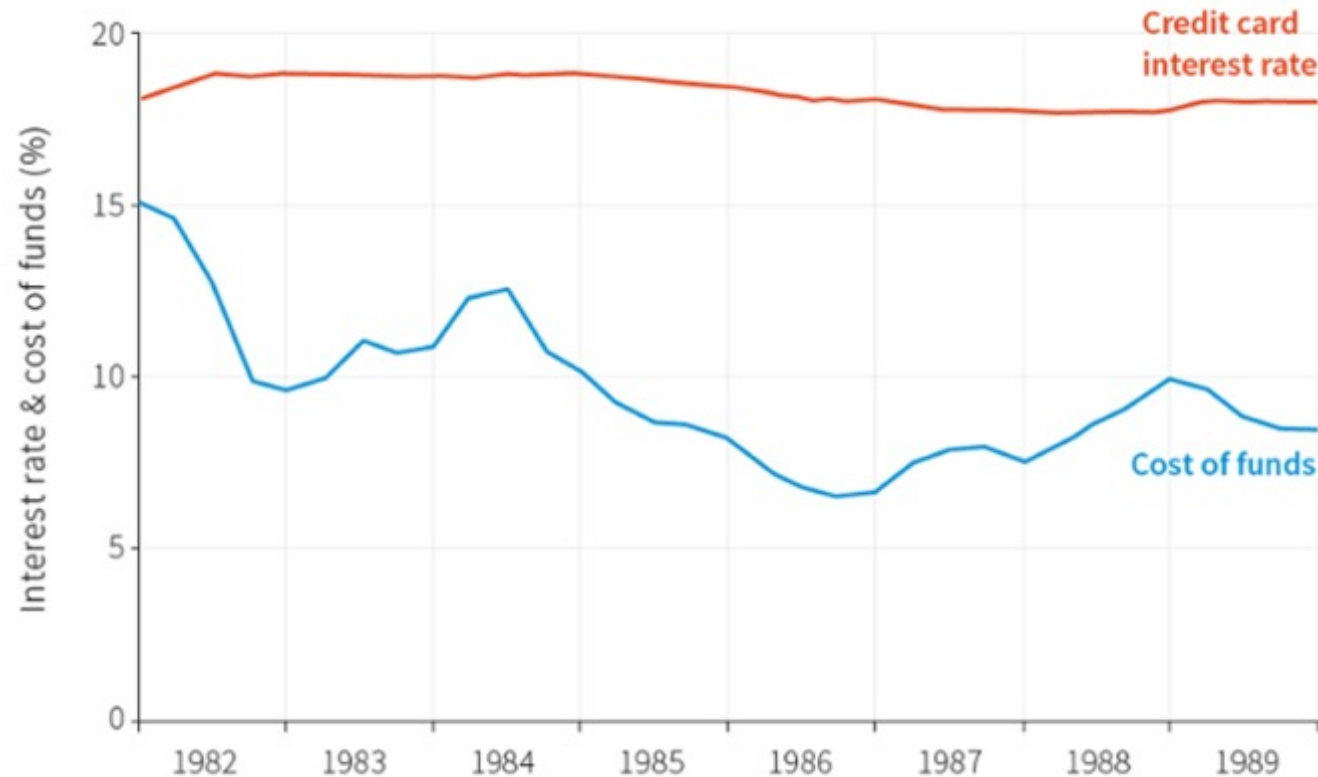
Is the model
of perfect
competition
a good
model?

8.9 Looking for competitive equilibria

Economists rely on two tests for a competitive equilibrium

Do firms sell goods at a price equal to marginal cost?

Do all trades take place at the same price?



The Hobbit: An Unexpected Journey	
SUPPLIER	PRICE INCLUDING POSTAGE (\$)
Game	14.99
Amazon UK	15.00
Tesco	15.00
Asda	15.00
Base.com	16.99
Play.com	17.79
Zavvi	17.95
The HUT	18.25
I want one of those	18.25
Hive.com	21.11
MovieMail.com	21.49
Blackwell	24.99

Fulton fish market

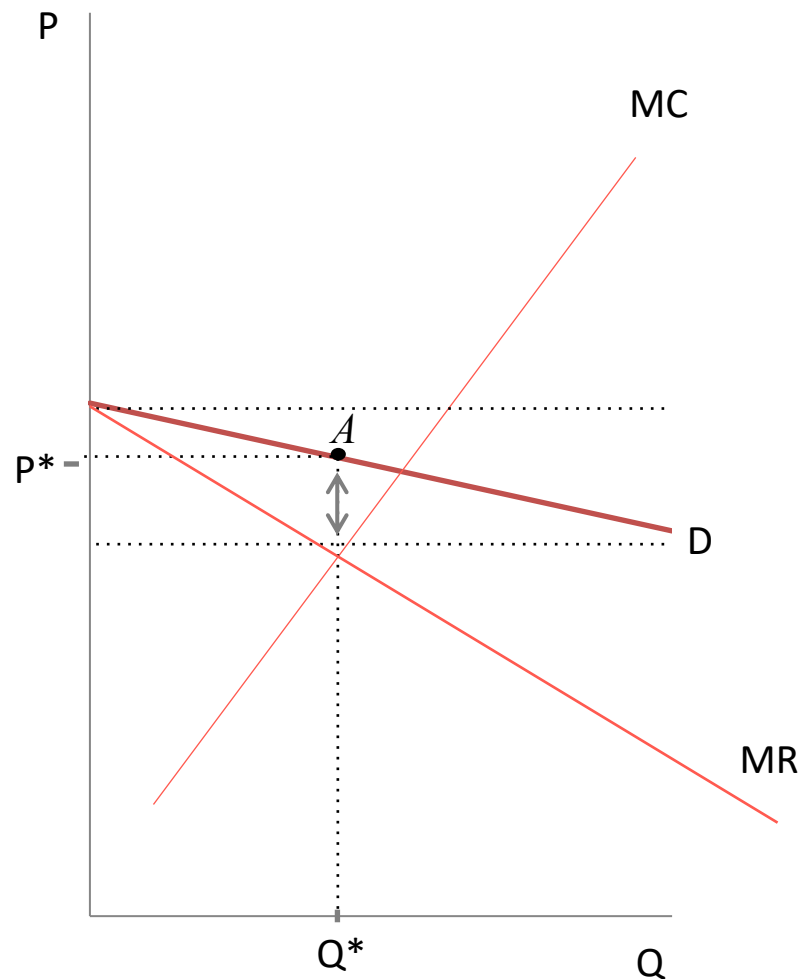


Economist in action *Kathryn Graddy*

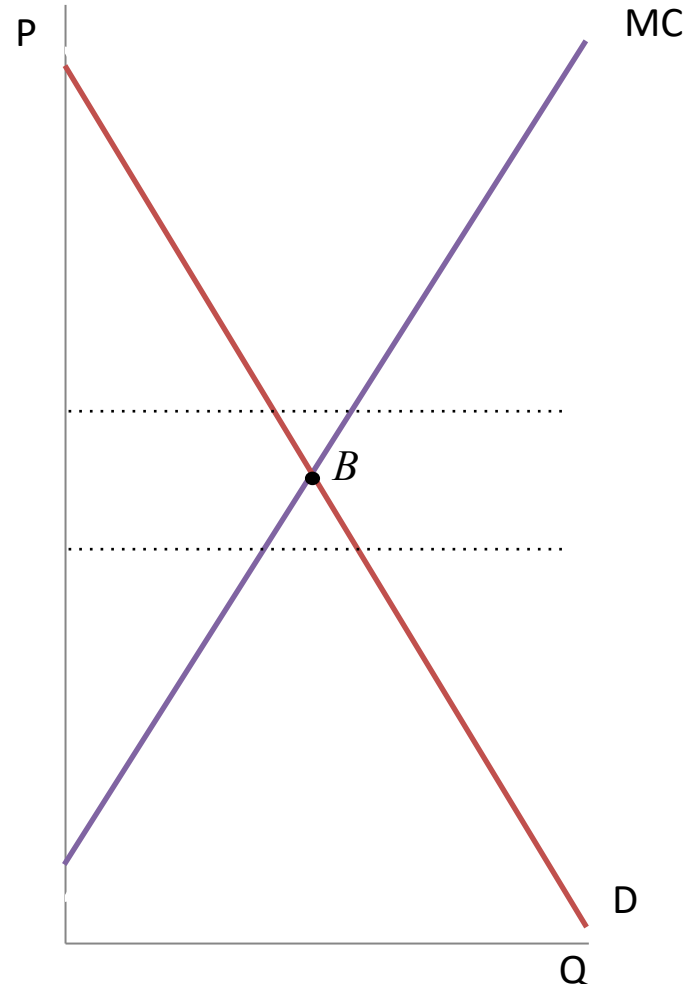
- Is the Fulton fish market perfectly competitive?
- Why or why not?
- Would it be more useful to model firms' behavior as price setters or price takers?

Market for Choccos and chocolate bars

Market for Choccos (produced by one firm)



Market for chocolate bars



- Many chocolate bars are close substitutes for Choccos.
- This competition means that Choccos has little market power:
 - Its demand curve is almost flat
 - It can barely raise the price by constraining production
- The firm chooses a price and quantity where $P \approx MC$
- Other firms do the same.
- The market supply curve approximately equal to the horizontal summation of all firms' MC curves.
- For many applications, modeling the market as competitive is reasonable.